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Semiconductor Capital Equipment | North America

2Q'26 WFE update, memory mismatch unresolved, LAM up to OW, AMAT down to EW

We revise up our 2026/27 WFE market forecast to +27%/+28% and introduce 2028 (+13%). We're positive about NAND and Intel into 2027. MKSI to Top Pick.

What's Changed? We revise up our 2026 forecast from \$143bn (+23% y/y) to \$149bn (+27%) and our 2027 forecast comes up from \$182bn (+27%) to \$191bn (+28%). For 2026, we revise up DRAM/foundry logic by \$3bn/\$2bn, while for 2027 we revise up DRAM/NAND by \$5bn/\$4bn. **We introduce our 2028 WFE forecast of \$215bn (+13% y/y).** We model 18% growth in foundry logic and 6% growth in memory. Our foundry logic growth forecast is led by a broadening of spenders, while in memory we would look to adjust our estimates as we gain greater visibility into project scale. Overall, we believe the WFE cycle has longevity, with memory clean room availability constraining some near-term upside while extending the medium-term opportunity, alongside a broadening of leading-edge logic spending.

We have 4 key takeaways this quarter: **1) DRAM:** Our 2027 DRAM WFE forecast of \$57bn supports 31% bit growth, but with unconstrained bit demand at 40%+, we do not think the supply/demand imbalance is resolved. **2) NAND:** We expect NAND WFE to be the fastest-growing end-market in 2027, up 52% y/y, as greenfield activity accelerates across key producers. Similar to DRAM, we see unconstrained bit demand at 40%+, vs our 2027 bit supply growth forecast of 32%. **3) Intel:** We see capex increases as inevitable, even in a scenario where Intel remains an IDM. We estimate that 5mn incremental Xeon 6s would require \$5.3bn+ of WFE, or a minimum of \$24bn if Intel aims to capture two-thirds of the incremental 33mn-unit CPU TAM by 2030. **4) Packaging:** We forecast packaging to grow ahead of WFE this year, at 41% vs 27%. However, we see the strongest value proposition at the front end, benefiting AMAT, LAM, KLA, and TEL, as front-end packaging grows 50% y/y and increases to 50% of the packaging TAM. Our broader 2027 WFE pecking order is NAND > non-TSMC foundry > TSMC > DRAM.

LAM up to OW, AMAT down to EW. We've preferred DRAM WFE exposure over NAND WFE given the relative revisions, and hence have preferred AMAT over LAM. The magnitude of our DRAM WFE revisions have narrowed, we're more positive on NAND WFE revisions from here. We revise up our CY27 forecast for LAM from \$34.6bn/\$9.46 to \$35.4bn/\$9.71 as we now model 59% NAND systems growth, surpassing prior peak in 2021. We had valued LAM at 31x vs AMAT at 28x, approx. a 10% premium to reflect our confidence on AMAT's 2026 share gains. Given our confidence on LAM's 2027 share gains, we raise that premium to 20%, a touch-above the 3-year average of 16%, which brings up our target multiple to 34x and raises our price target from \$293 to \$331 implying 16% upside.

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SEMICONDUCTOR CAPITAL EQUIPMENT		
North America		
Industry View	In-Line	
WHAT'S CHANGED		
Lam Research Corp (LRCX.O)	From	To
Rating	Equal-weight	Overweight
Price Target	\$293.00	\$331.00
MKS Inc. (MKSI.O)	From	To
Price Target	\$354.00	\$374.00
Top Pick	AMAT.O	MKSI.O
Applied Materials Inc. (AMAT.O)	From	To
Rating	Overweight	Equal-weight
Top Pick	AMAT.O	MKSI.O

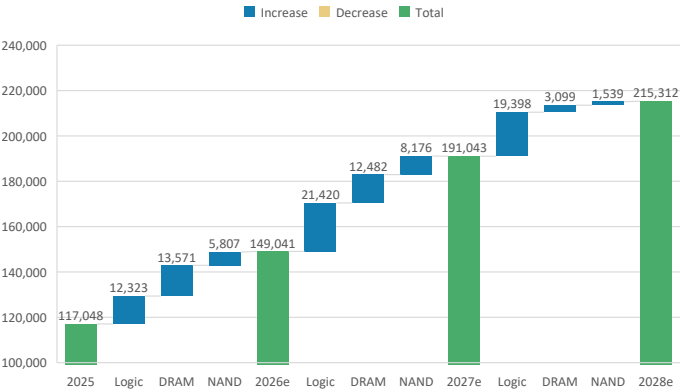
MKS Inc. Riding the NAND wave, Raise to Top Pick

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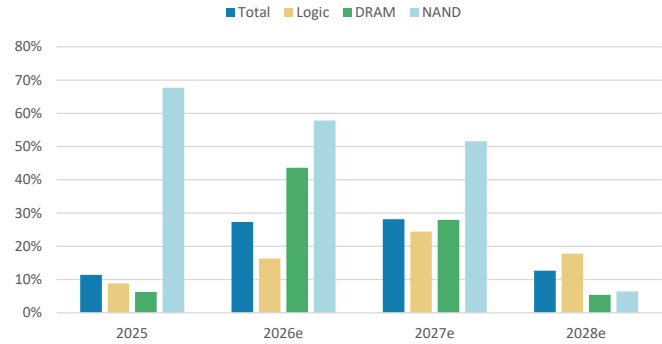
Our story in 6 charts

Exhibit 1: Our 2026 revisions are led by DRAM/Logic and 2027 led by DRAM/NAND



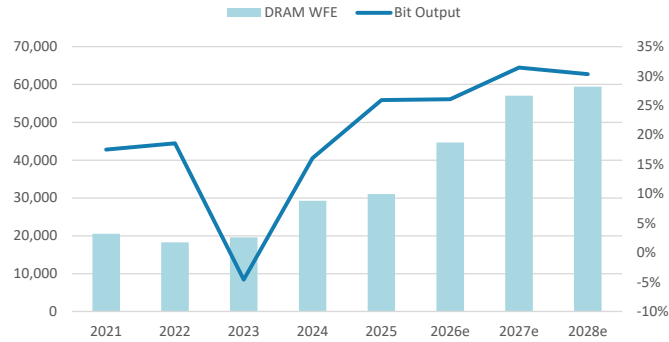
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 2: Y/Y growth will be led by NAND, partially due to a low base, but we're positive on NAND WFE



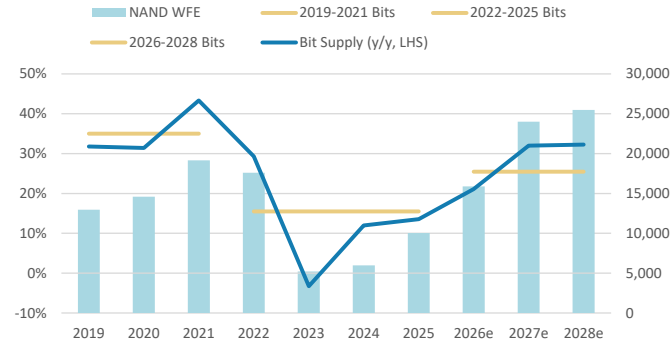
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 3: Our 2027 DRAM WFE forecast of \$57bn supplies 31% bit growth, below where we think unconstrained bit demand is (40%+)



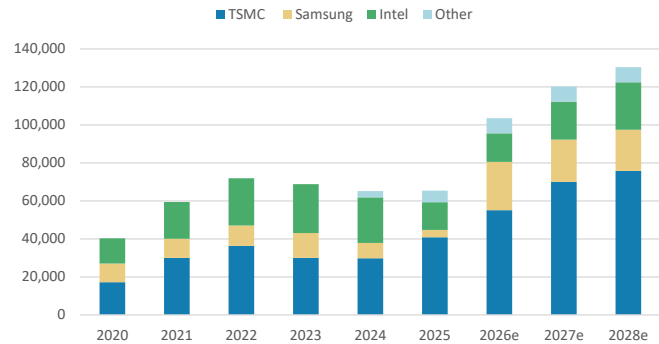
Source: TrendForce, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 4: Our 2027 NAND WFE forecast of \$24bn supplies 32% bit growth, below where we think unconstrained bit demand is (40%+)



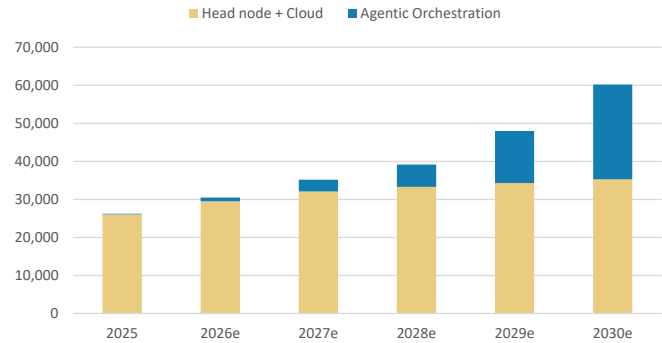
Source: TrendForce, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 5: We model leading edge logic capex growth of 58% in 2026



Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 6: If Intel wants to build supply to capture two-thirds of incremental CPU units by 2030, we think that would require \$23.8bn+ of WFE



Source: Morgan Stanley Research estimates

WFE Forecast Table

Exhibit 7:

(\$mn)	2021	2022	2023	2024	2025	2026e	2027e	2028e
WFE Revenue	91,092	96,222	96,008	105,057	117,048	149,041	191,043	215,312
Semi Revenue	553,540	573,978	526,820	630,549	791,061	1,513,890	1,835,394	1,835,394
Semi Capex	157,788	187,982	176,924	198,540	215,972	347,044	377,535	368,449
Y/Y Change %								
WFE Revenue	43%	6%	0%	9%	11%	27%	28%	13%
Semi Revenue	26%	4%	-8%	20%	25%	91%	21%	0%
Semi Capex	33%	19%	-6%	12%	9%	61%	9%	-2%
Metrics								
WFE Intensity	16%	17%	18%	17%	15%	10%	10%	12%
WFE% of Total Capex	58%	51%	54%	53%	54%	43%	51%	58%
Semi Capital Intensity	29%	33%	34%	31%	27%	23%	21%	20%
3rd Party Sources								
Semi	87,499	94,100	95,610	104,270	115,700	126,100	135,200	
Gartner	92,843	101,101	102,820	111,777	124,452	144,886	161,086	154,674
WFE by segment								
Foundry/Logic	50,265	59,180	70,023	69,192	75,307	87,631	109,051	128,449
Memory	39,828	35,854	24,794	35,228	41,118	60,496	81,153	85,792
DRAM	20,664	18,256	19,579	29,244	31,083	44,654	57,135	60,235
NAND	19,164	17,598	5,215	5,985	10,035	15,842	24,018	25,557
Other	999	1,188	1,192	637	622	914	839	1,071
Y/Y Change %								
Foundry/Logic	45%	18%	18%	-1%	9%	16%	24%	18%
Memory	42%	-10%	-31%	42%	17%	47%	34%	6%
DRAM	53%	-12%	7%	49%	6%	44%	28%	5%
NAND	31%	-8%	-70%	15%	68%	58%	52%	6%
Other	19%	19%	0%	-47%	-2%	47%	-8%	28%
% of Total								
Foundry/Logic	55%	62%	73%	66%	64%	59%	57%	60%
Memory	44%	37%	26%	34%	35%	41%	42%	40%
DRAM	23%	19%	20%	28%	27%	30%	30%	28%
NAND	21%	18%	5%	6%	9%	11%	13%	12%
Other	1%	1%	1%	1%	1%	1%	0%	0%
WFE Intensity								
Semi ex-Memory revenue	399,706	444,153	434,511	465,032	568,550	706,738	802,096	802,096
Memory revenue	153,834	129,825	92,309	165,516	222,512	807,153	1,033,298	1,033,298
DRAM	92,960	77,769	51,945	94,860	150,598	511,693	585,155	585,155
NAND	55,953	47,109	36,104	66,427	67,685	291,230	443,913	443,913
WFE Intensity								
Foundry/Logic	13%	13%	16%	15%	13%	12%	14%	16%
Memory	26%	28%	27%	21%	18%	7%	8%	8%
DRAM	22%	23%	38%	31%	21%	9%	10%	10%
NAND	34%	37%	14%	9%	15%	5%	5%	6%
% of Memory WFE								
DRAM	52%	51%	79%	83%	76%	74%	70%	70%
NAND	48%	49%	21%	17%	24%	26%	30%	30%

Source: Gartner, SEMI, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 8:

(\$mn)	2021	2022	2023	2024	2025	2026e	2027e	2028e
WFE by Region								
Total	91,092	96,222	96,008	105,057	117,048	149,041	191,043	215,312
North America	7,755	10,239	11,346	13,657	10,534	12,221	15,666	17,656
Europe	2,879	5,982	5,741	4,202	2,341	2,534	2,866	3,230
Japan	6,879	7,244	6,438	7,354	8,779	10,433	13,755	15,718
Korea	22,757	19,655	18,965	18,910	23,410	33,534	44,513	50,168
Taiwan	23,566	26,566	19,501	15,233	27,506	38,751	53,492	60,287
China	24,695	22,105	31,682	42,548	41,552	48,140	56,740	64,594
Other	2,561	4,431	2,334	3,152	2,926	3,428	4,012	3,660
WFE by Region (Y/Y)								
North America	28%	32%	11%	20%	-23%	16%	28%	13%
Europe	8%	108%	-4%	-27%	-44%	8%	13%	13%
Japan	14%	5%	-11%	14%	19%	19%	32%	14%
Korea	51%	-14%	-4%	0%	24%	43%	33%	13%
Taiwan	54%	13%	-27%	-22%	81%	41%	38%	13%
China	48%	-10%	43%	34%	-2%	16%	18%	14%
Other	40%	73%	-47%	35%	-7%	17%	17%	-9%
WFE by Region (% of Total)								
North America	9%	11%	12%	13%	9%	8%	8%	8%
Europe	3%	6%	6%	4%	2%	2%	2%	2%
Japan	8%	8%	7%	7%	8%	7%	7%	7%
Korea	25%	20%	20%	18%	20%	23%	23%	23%
Taiwan	26%	28%	20%	15%	24%	26%	28%	28%
China	27%	23%	33%	41%	36%	32%	30%	30%
Other	3%	5%	2%	3%	3%	2%	2%	2%

Source: Gartner, SEMI, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

How do our views change (2026/2027)?

What's Changed? We revise up our 2026 forecast from \$14.3bn (up 23% y/y) to \$14.9bn (up 27%) and our 2027 forecast comes up from \$182bn (up 27%) to \$191bn (up 28%).

Our View: Our largest revision is in NAND, where we are materially increasing our 2027 greenfield wafer assumptions across Samsung and Kioxia.

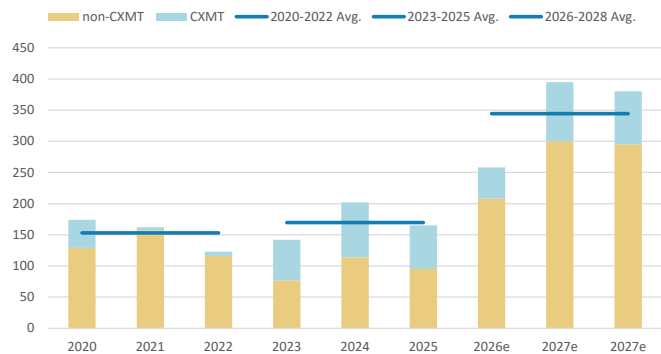
DRAM: capped at low-30% bit output?

We make 2 changes to our DRAM WFE forecasts which brings up our 2026/2027 DRAM WFE forecast from \$41.7bn/\$52.2bn to \$44.6bn/\$57.1bn:

- Greenfield: We adjust our 2026/2027 forecast from 278/355kwpm to 258/395kwpm as we tweak down our overly aggressive 2026 Hynix forecast while revising up 2027 forecasts across all manufacturers.
- Bit Output: Our revised wafer forecasts lead to our bit supply growth forecasts being revised up from 24%/29% to 26%/31%.

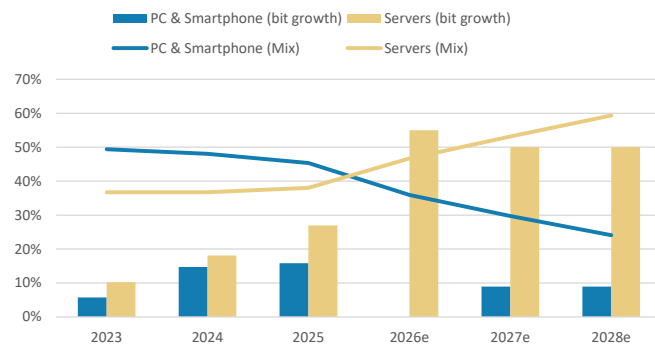
We aren't sure how much higher our forecasts can go; as it would imply extremely aggressive wafer start ramps at new fabs or a majority of ASML's EUVs going towards DRAM, straining logic in the process.

Exhibit 9: We now model annual DRAM greenfield adds to average 344kwpm during 2026-28, up from 170kwpm in 2023-25



Source: TrendForce, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

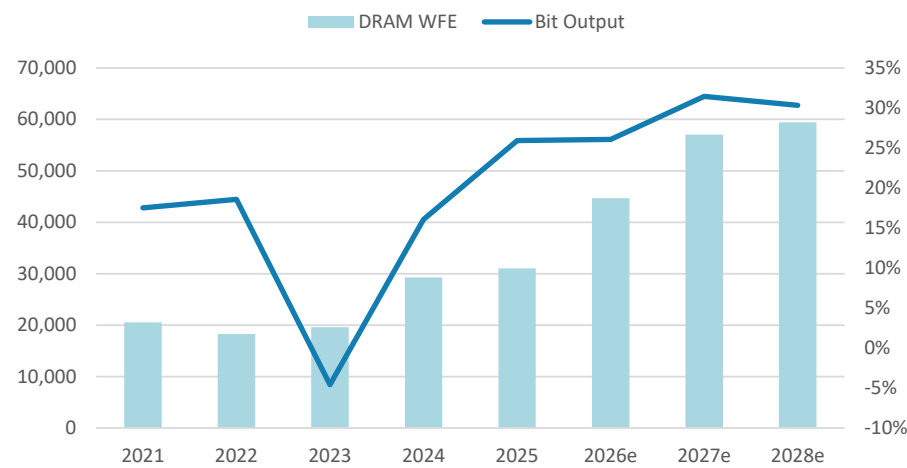
Exhibit 10: We model servers to be 59% of DRAM bits in 2028, up from 38% in 2025, and PC/Smartphone down from 45% to 24%



Source: TrendForce, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 11:

We model 31%/30% bit supply growth in 2027/2028, up from 12% during 2023-25 and 17% during 2020-22



Source: TrendForce, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

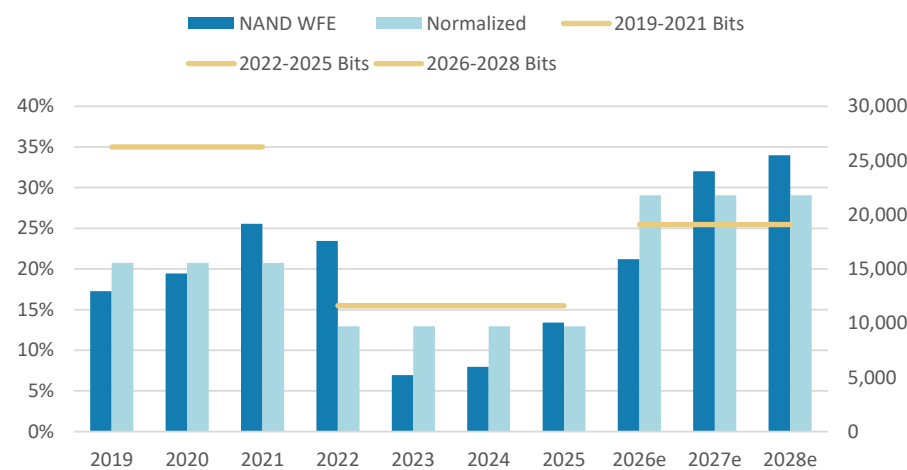
NAND: get ready for a NAND WFE acceleration

We make 2 changes to our NAND WFE forecasts which brings up our 2026/2027 NAND WFE forecast from \$15.4bn/\$19.7bn to \$15.8bn/\$24.0bn:

- Greenfield: We adjust our 2026/2027 forecast from 105/125kwpm to 95/145kwpm as we tweak down our 2026 Samsung forecast while revising up 2027 forecasts at Samsung and Kioxia.
- Bit Output: Our revised wafer forecasts lead to our bit supply growth forecasts being revised up from 20%/30% to 21%/32%.

We're confident on NAND WFE accelerating into 2027.

Exhibit 12: We model NAND bit supply growth to increase from 17% in 2026 to 33% in 2028



Source: TrendForce, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

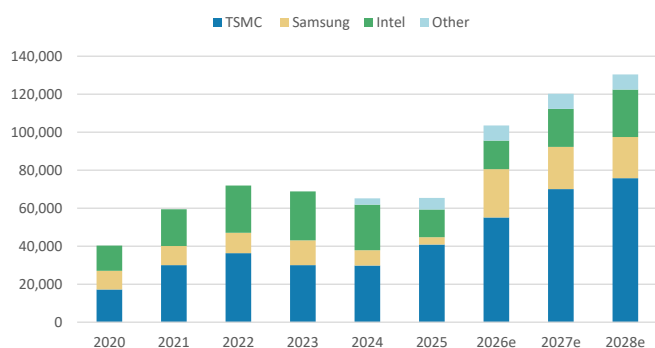
Logic: Broadening of spenders

We fine tune our 2026-27 WFE forecasts from \$85.4bn/\$109.4bn to \$87.6bn/\$109.0bn. As we think about further revisions from here, we think upside is skewed towards 2 areas:

- Intel: We think there is an inevitability towards Intel raising capex on the back of CPU demand, and we see further revisions down the line if the company captures a material foundry customer or sees some sort of development with TeraFab.
- Analog/MCU: Analog/MCU companies spoke of increasing utilization rates across the board at their MarQ reports. NXP spoke of front-end utilization coming up from the low 80s in the 1H to mid 80s in the 2H, ON spoke about manufacturing utilization increased sequentially to 77% and expecting Q2 flat to up slightly, and Infineon mentioned 300mm fabs really have a high capacity utilization. We don't see an urgency to increase capex amongst these players, nor do we think capex will materially pick up enough to shift overall WFE growth rates. However, we do believe we can start modelling positive y/y growth across non-China ICAPS.

We aren't quite ready to put that upside fully into numbers, but we think risk is skewed to the upside.

Exhibit 13: We model leading edge logic capex growth of 58% in 2026



Source: Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 14: We estimate that 22.4mn incremental Server CPUs would require WFE of \$23.8bn+ from Intel

WFE per CPU Needs (\$mn)	Intel Xeon 6	Intel Xeon 7
Total	23,854	27,913
Logic	23,517	27,577
2nm	0	20,311
3nm	20,164	0
7nm	3,354	7,266
EMIB	336	336

Wafers per CPU Needs	Intel Xeon 6	Intel Xeon 7
Total	105,182	108,441
Logic	100,698	103,957
2nm	0	63,588
3nm	82,066	0
7nm	18,632	40,368
EMIB	4,484	4,484

Source: Morgan Stanley Research estimates

Introducing 2028: Longevity of Demand

Our View: We believe in the longevity of this WFE cycle led by AI. We expect a broadening of foundry logic spending to continue to play out in 2028, while for memory we are looking to raise numbers as fab timelines become clearer.

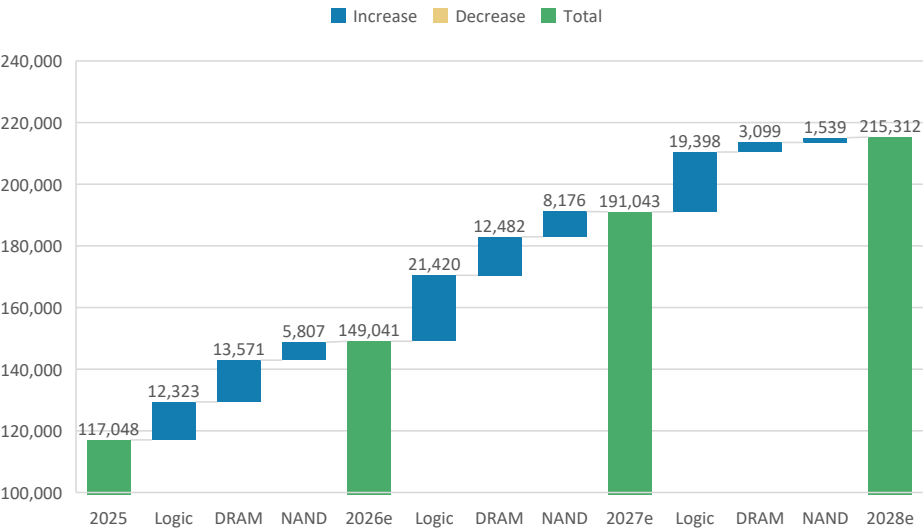
The Numbers: We model total WFE to be up 13% in 2028, led by foundry logic up 18% y/y vs memory up 6%.

Our thoughts by end-markets are as follows;

- Logic \$128.4bn (+18% y/y): We model leading edge capex to increase from TSMC & Intel as well as trailing edge capex to gradually recover led by Analog/MCU players across the board.
- DRAM \$60.2bn (+5% y/y): We model greenfield to remain roughly flat at 380kwpm (2027 395kwpm) as we await more clarity around the scale of MU's PSMC P5 fab and CXMT's Shanghai fab.
- NAND \$25.6bn (+6% y/y): We model greenfield to accelerate from 145kwpm to 170kwpm led by all players. We see risk skewed to the upside especially from MU's Singapore fab (Fab 10(B)).

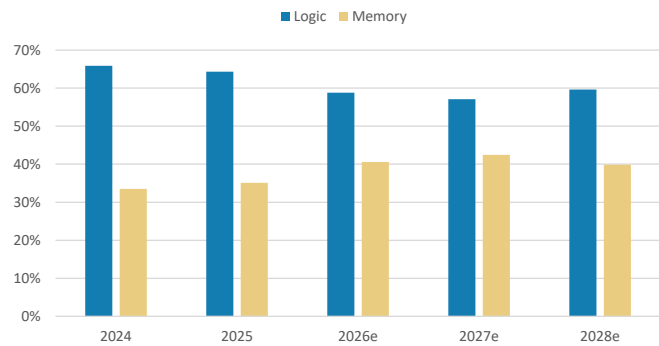
Across DRAM/NAND we would look to adjust numbers as we get a better understanding of project scale.

Exhibit 15: We model logic to drive 2028 growth, but risk is skewed to the upside for memory



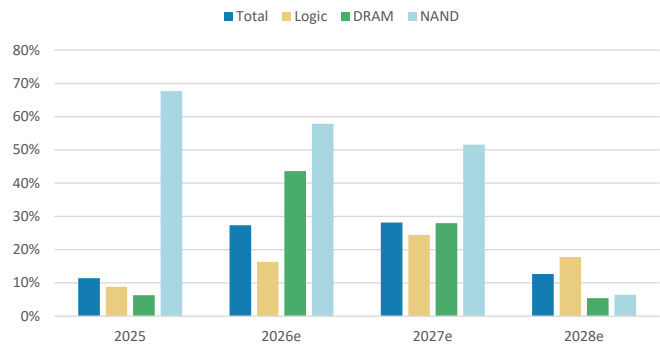
Source: Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 16: We model logic/memory mix to narrow from 65%/35% in 2025 to 58%/42% in 2027



Source: Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 17: Y/Y growth will be led by NAND, partially due to a low base



Source: Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Thought #1: NAND WFE

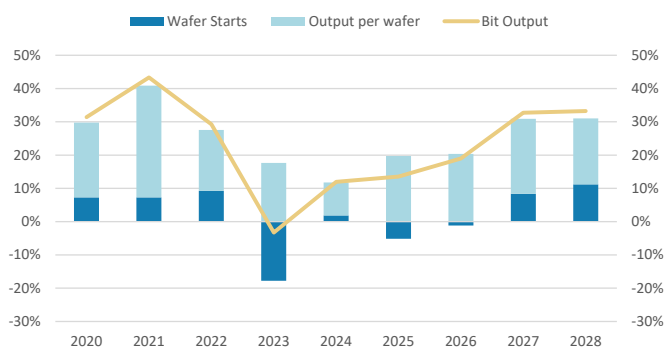
Our View: We think NAND WFE surpasses historic highs in 2027 on the back of increased clean room allocation and lithography constraints for DRAM.

The Numbers: We raise our 2027 NAND WFE forecast from \$19.7bn to \$24.0bn, above prior high of \$19.1bn in 2021.

NAND WFE has continued to stay low. NAND is a cyclical industry, and past cycles have always seen spending resume at some point. Our view is predicated on history and bit growth, and see spending reaccelerating, with the debate being “When” not “If”. However, NAND WFE shipments have tracked at historic lows during Q3 25 to Q1 26, and our thesis has been slow to play out for 3 reasons:

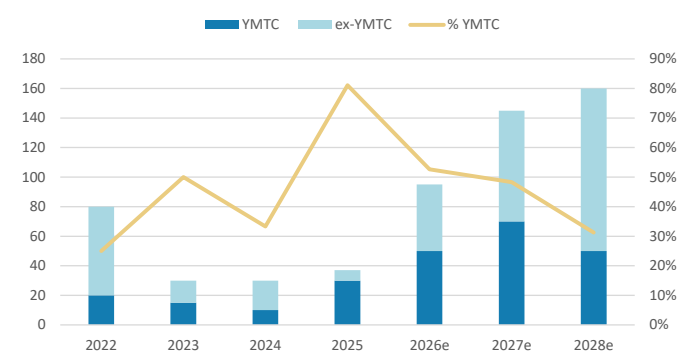
- **Clean room availability:** DRAM WFE has accelerated from strength to strength as Samsung and Hynix have allocated all incremental clean room towards DRAM as well as shifting clean room initially allocated from NAND to DRAM.
- **Capital Discipline:** Samsung, Hynix and Micron have shifted their capex budgets away from DRAM and towards NAND, while Kioxia has been more capital disciplined, leaving just YMTC to add NAND wafers.
- **Bit Demand:** We estimate node upgrades have been able to satisfy high-teens bit growth, which is what the demand profile of the industry has been during 2023-2025.

Exhibit 18: NAND supply growth during 2023-2025 has largely been supplied by upgrades (more bits per wafer)



Source: TrendForce, Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 19: As a result, greenfield in the industry was minimal



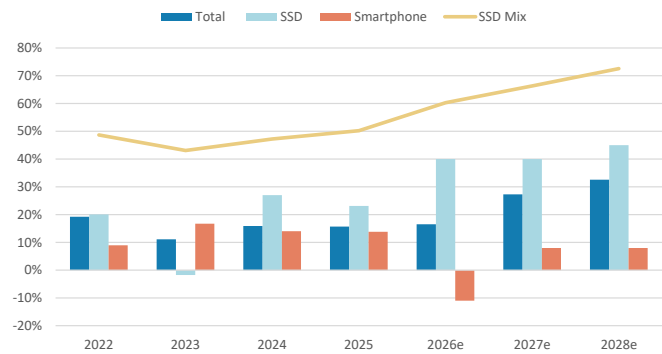
Source: TrendForce, Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

We think NAND WFE accelerates from here for 3 reasons:

- **Clean room availability:** as more memory clean room capacity comes online while EUV supply becomes incrementally tighter, we think NAND receives increased allocation.
- **Capital Discipline:** SNDK guided JunQ GM% to 80%, making NAND far more profitable relative to 9 months ago (SepQ 2025 GM 30%). Sandisk also disclosed \$42bn in RPO, with \$11bn in financial guarantees, giving line of sight to profitability on incremental bit production.

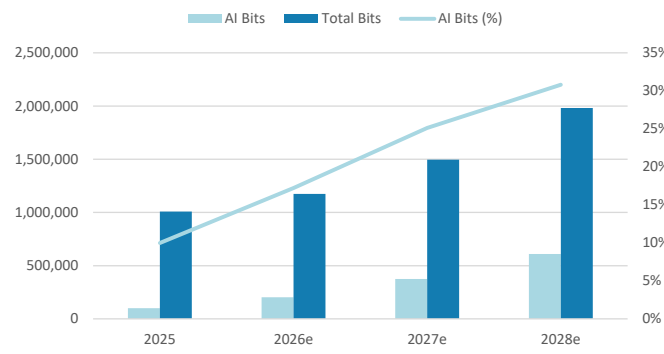
- **Bit Demand:** We think unconstrained bit demand is 40%+ and we think LTAs necessitate spending to a certain extent.

Exhibit 20: We think unconstrained bit demand is 40%+, largely driven by server demand (SSDs)



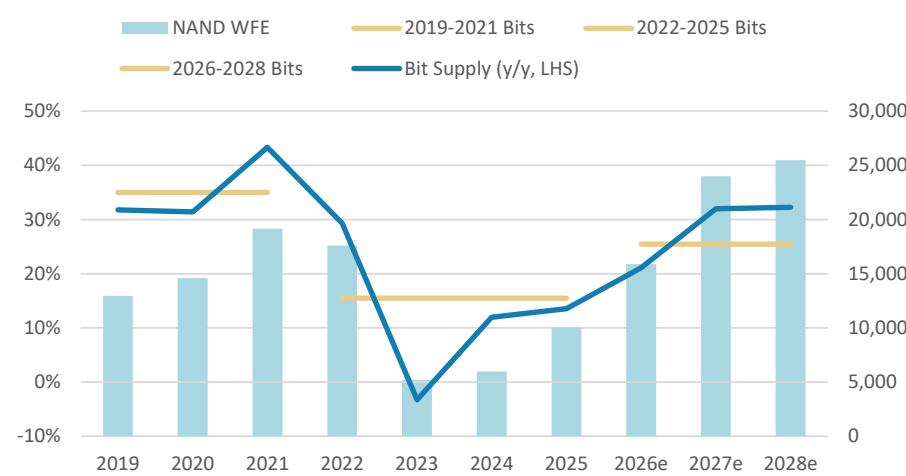
Source: TrendForce, Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 21: We model server as a % of NAND bits to reach 73% in 2028 driven by AI bits reaching 31%



Source: TrendForce, Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 22:
We think NAND WFE increases to \$24bn in 2027, above prior peak of \$19bn in 2021.



Source: TrendForce, Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

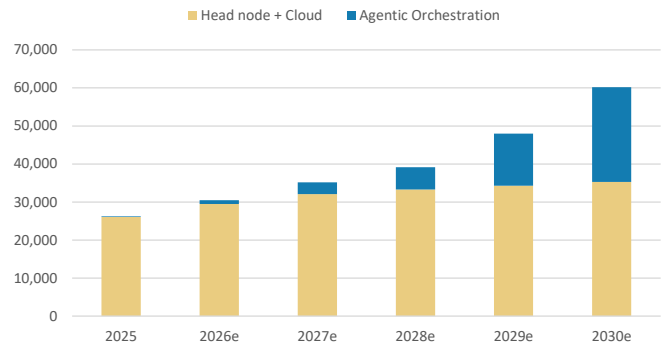
Thought #2: Intel

Our View: We think rising CPU demand makes an Intel capex hike for 2026/2027 inevitable, with foundry customer acquisitions making capex revisions heavily skewed to the upside.

The Numbers: We model Intel capex of \$20bn/\$25bn in 2026/27 but with risk heavily skewed to the upside given 5mn CPU packages (Xeon 6) per year of incremental production necessitates approx. 22kwpm logic wafers/\$5.3bn of WFE.

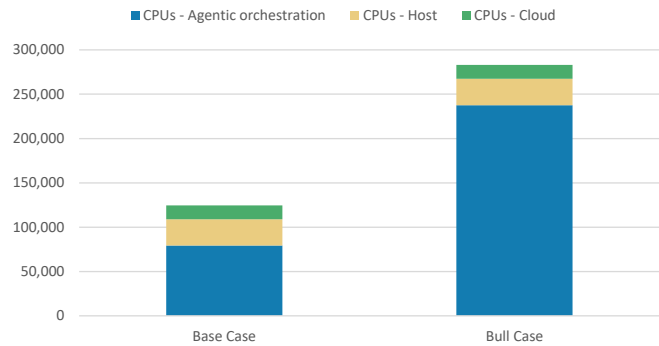
\$125bn 2030 Server CPU Base Case. Our global semis team recently raised their 2030 CPU TAM by 25% to US\$125bn, driven by a US\$79bn orchestration CPU layer on top of US\$45bn host/cloud CPU TAM; their top-down bull case reaches US\$283bn as dedicated agentic CPU racks scale. We model server CPUs to grow from 26.2mn units in 2025 to 60.1mn in 2030. For further details see: [Global Technology: Agentic AI – The Surge Begins \(11 May 2026\)](#)

Exhibit 23: We model server CPUs to grow from 26.2mn units in 2025 to 60.1mn in 2030



Source: Morgan Stanley Research estimates

Exhibit 24: Our bull case implies a \$238bn CPU orchestration TAM.



Source: Morgan Stanley Research estimates

Inevitability to Intel Capex revisions. Were Intel to take part in this CPU upcycle, we think there is an inevitability to capex hikes. We estimate that if the company wants to produce 5mn incremental Xeon 6s, the company will have to spend \$5.3bn in WFE. Assuming Intel wants to take two-thirds of the incremental CPU units, we think the company will have to expand logic wafer capacity by a minimum of \$23.8bn. Assuming 2 compute tiles for Xeon 7, we estimate Intel will have to spend \$27.9bn in WFE. We see capex tracking possibly even higher:

- **Intel Foundry:** Our capex argument is purely based on Intel looking to increase capex to meet rising CPU demand, i.e. Intel IDM. Foundry customers acquisition will necessitate spending beyond the \$23.8-27.9bn of WFE to meet rising CPU demand.

Exhibit 25: We estimate 5mn incremental server CPUs would require \$5.3bn+ of WFE

WFE per CPU Needs (\$mn)	Intel Xeon 6	Intel Xeon 7
Total	5,326	6,233
Logic	5,251	6,158
2nm	0	4,535
3nm	4,502	0
7nm	749	1,623
EMIB	75	75

Wafers per CPU Needs		
Total	23,487	24,214
Logic	22,485	23,213
2nm	0	14,199
3nm	18,325	0
7nm	4,160	9,014
EMIB	1,001	1,001

Source: Morgan Stanley Research estimates. 2nm = Intel 18A, 3nm = Intel 3.

Exhibit 26: Two-thirds of the incremental units through 2030 (22.4mn units) would require \$23.8bn+ of WFE

WFE per CPU Needs (\$mn)	Intel Xeon 6	Intel Xeon 7
Total	23,854	27,913
Logic	23,517	27,577
2nm	0	20,311
3nm	20,164	0
7nm	3,354	7,266
EMIB	336	336

Wafers per CPU Needs		
Total	105,182	108,441
Logic	100,698	103,957
2nm	0	63,588
3nm	82,066	0
7nm	18,632	40,368
EMIB	4,484	4,484

Source: Morgan Stanley Research estimates. 2nm = Intel 18A, 3nm = Intel 3.

KLA an outsized beneficiary. Intel has not been a 10% customer for KLA since F6/15, but we think Intel is leaning on KLA as a key partner to improve yield in order to improve margin and become a viable foundry. While key Intel suppliers like ASML and Tokyo Electron will benefit from an Intel capex resurgence, we think KLA stands as an outsized beneficiary on the back of increased process control intensity and share gain.

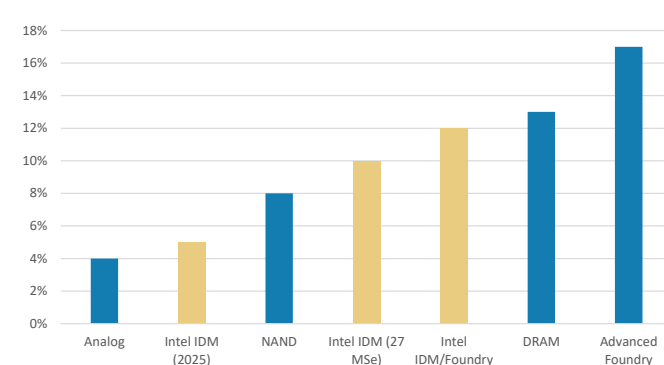
- Intel Foundry Direct Connect (April 29th 2025): "The second pillar is quality. Intel has to make significant improvements in ensuring that customers have no negative impact from our quality based deviations. We have to change from a test based quality detection *back to an in-line quality detection, and we are investing heavily in this area.* Not only through test chips but by *partnering with our equipment suppliers to install all the in-line detection capabilities (1:09.22-1:09.45)*" -Naga Chandrasekaran (GM, Intel Foundry Technology Manufacturing).
- MarQ 2026 Earnings (April 23rd 2026): "we've brought in external partners that are particularly good at, including metrology, that's helped us execute better there (yield) and we're starting to see the benefits of that this quarter."

Exhibit 27: KLA can 6.4x their Intel revenue from 2025 to 2027 on the back of PC intensity and share gain

(\$mn)	2025 MSe IDM	2027 MSe IDM	2027 MSe IDM/Foundry
Capex	14,646	20,000	25,000
WFE	7,323	12,000	15,000
WFE Mix (%)	50%	60%	60%
Process Control (%)	5%	10%	12%
KLA Share (%)	40%	60%	60%
KLA Revenue	146	720	1,080
KLA % of WFE	2.0%	6.0%	7.2%

Source: Morgan Stanley Research estimates.

Exhibit 28: We expect Intel's process control intensity to reach 12% in an IDM/Foundry scenario



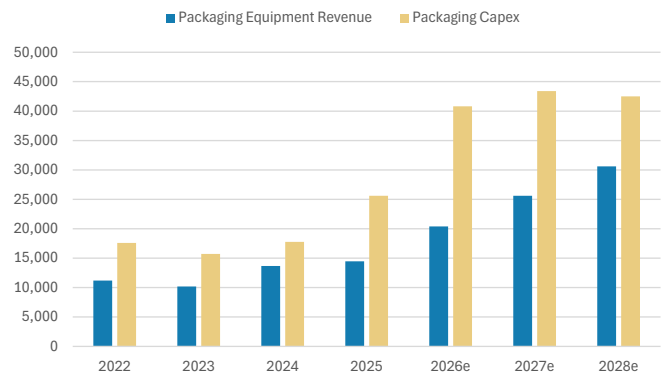
Source: Company Data, Morgan Stanley Research estimates.

Thought #3: Advanced Packaging

Our View: We forecast packaging to grow ahead of WFE this year but the value proposition is at the front-end players rather than the traditional back-end players.

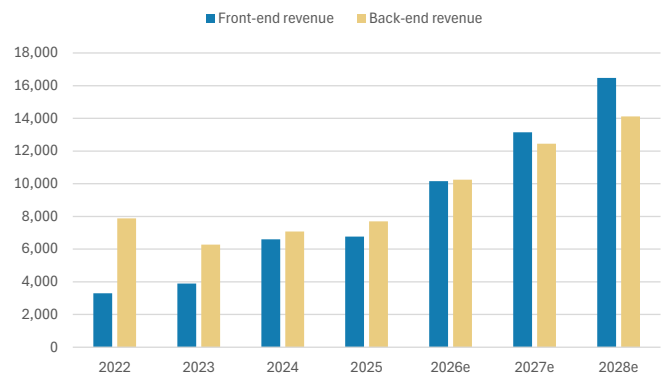
The Numbers: We model the broader packaging market to grow 41%/25% in 2026/2027 but with the value proposition shifting to the front-end as front-end packaging revenue grows 50% y/y in 2026 and increases to 50% of packaging.

Exhibit 29: We model packaging equipment to grow 41%/25% in 2026/2027...



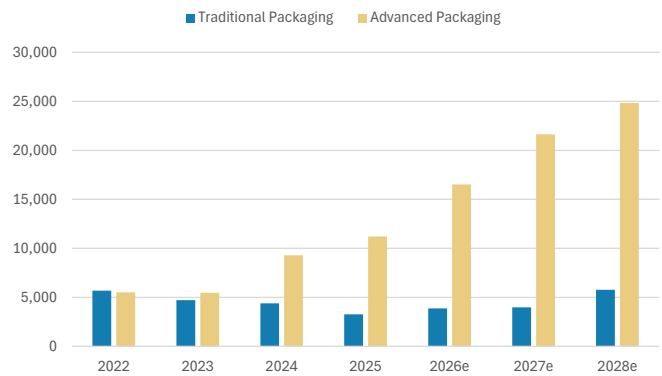
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 31: An increasing portion of packaging revenue is captured by the front-end players (AMAT/LAM/KLA/TEL)...



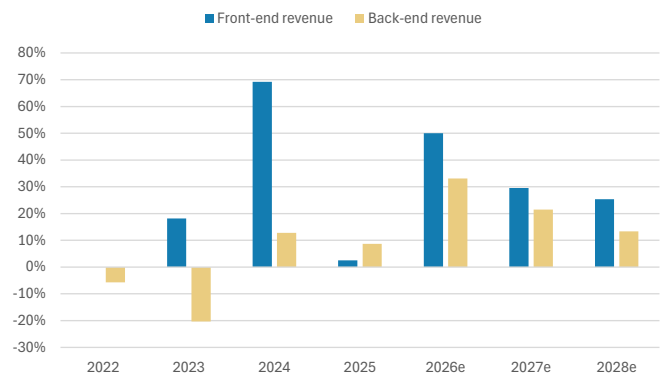
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 30: ...largely driven by Advanced Packaging (HBM, CoWoS, SoIC)



Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 32: ...and we forecast a higher growth rate vs the front-end cos through 2028



Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

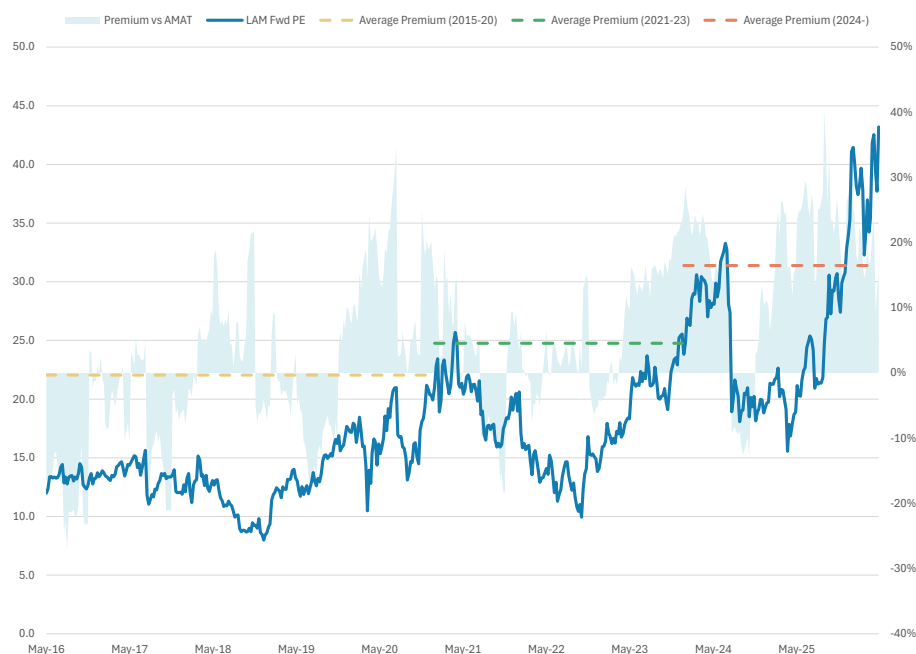
LAM up to OW, AMAT down to EW

LAM's valuation premium vs AMAT has tracked in 3 stages over the last 10 years:

- 0% (2015-20): minimal share movement between the 2 companies.
- 5% (2021-23): minimal share shift during 2021/22, initial premium during 2023 due to depressed NAND shipments
- 16% (2024-): LAM share gains vs AMAT share loss due to LAM's share gains vs AMAT's share loss.

We have argued for a narrower valuation gap between the 2 companies (10%) as we were confident AMAT would be a share gainer in 2026. As we look towards 2027, LAM's share gain prospects look stronger than AMAT due to the tailwinds in NAND. To reflect this view we expand our premium for LAM vs AMAT from 10% to 20%.

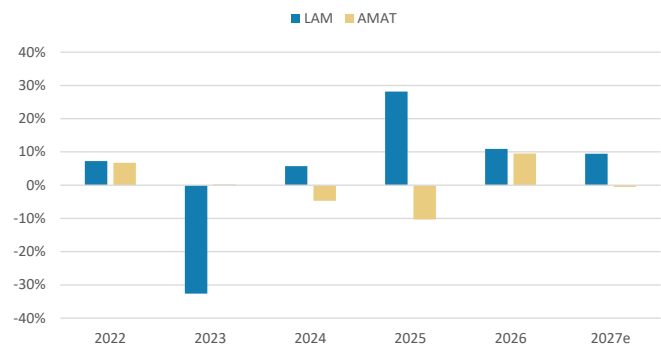
Exhibit 33: LAM FwdPE vs AMAT



Source: FactSet, Morgan Stanley Research

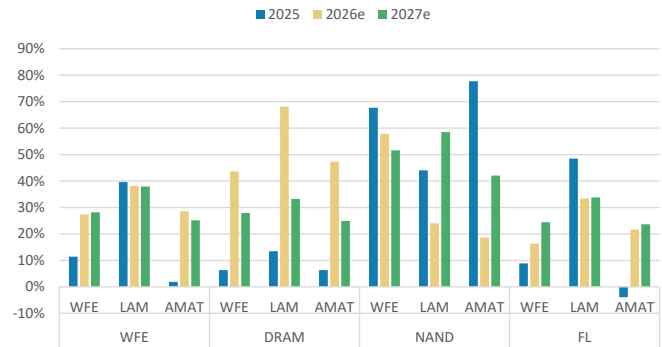
NAND shifts the needle for LAM, not for AMAT. We preferred AMAT in 2026 given the company's outsized exposure to DRAM. DRAM WFE upward revisions had a larger impact on AMAT than LAM (CY26 mix 32% vs 23%). As we look towards 2027, the growth rates of each company's NAND shipments aren't entirely that different, NAND WFE revisions will have a larger impact on LAM than AMAT due to mix (20% vs 7%). **Where could we be wrong?** 2 areas of risk for LAM are: 1) China: LAM has larger exposure to China (CY26e 32% vs AMAT 24%), meaning any tightening of restrictions will have an outsized impact on the company. 2) NAND: We haven't seen large scale NAND greenfield ex-China since 2022. While LAM had the incumbency in brownfield spending, given they are the existing installed base, incumbency doesn't matter as much for greenfield, especially in a new node.

Exhibit 34: We model LAM to see another year of share gain in 2027



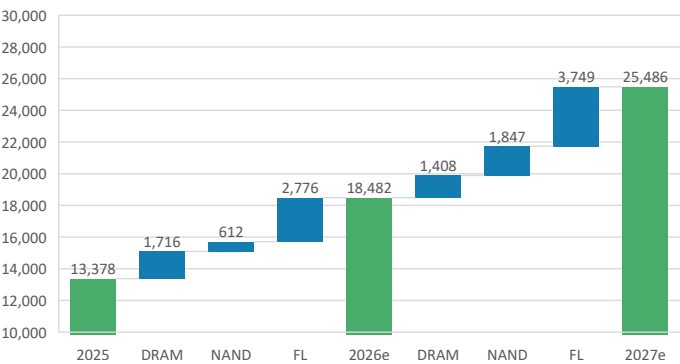
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 35: We expect share gains to persist in Foundry Logic



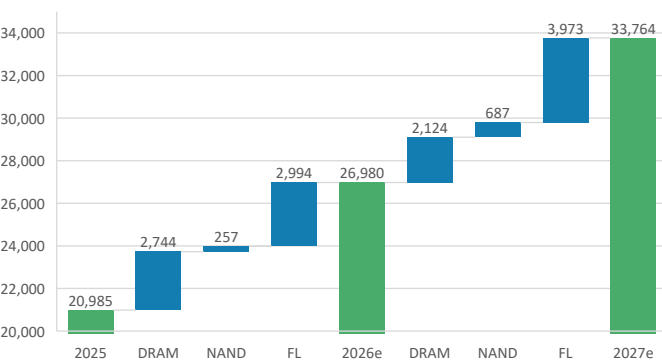
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 36: NAND helps shift the needle for LAM



Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 37: NAND doesn't shift the needle for AMAT



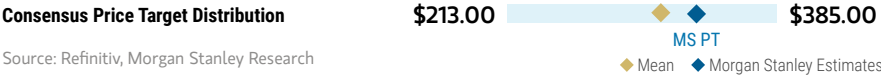
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Risk Reward – Lam Research Corp (LRCX.O)

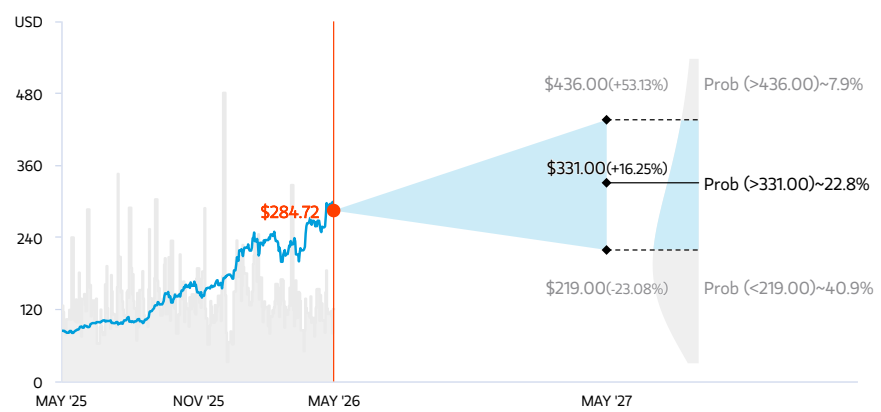
WFE outperformance intact with NAND & Logic

PRICE TARGET \$331.00

34x CY27 non-GAAP EPS of \$9.73, multiple reflects a 20% premium to AMAT (28x) to reflect LAM's market share gains.



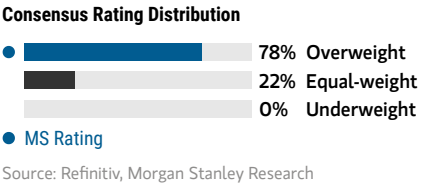
RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)



Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 15 May 2026. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology [here](#)

OVERWEIGHT THESIS

- After shipments underperformed WFE by 33ppt in 2023 due to a cyclical decline in NAND and China entity list impact, LAM outperformed WFE by 8ppt in 2024 and we estimate another 27ppt outperformance in 2025 driven by NAND, China, and TSMC. We model LAM's system shipments to outperform WFE in 2026 & 2027 for the 3rd and 4th consecutive year as NAND and non-China foundry logic continues to strengthen.
- LAM has unique drivers supporting the long-term growth story as the company is seeing gross margin expansion from shifting manufacturing to Malaysia, is positioned well to gain share with the 4F2 transition in DRAM and in advanced packaging.



Risk Reward Themes

New Data Era: Positive
Secular Growth: Positive
View descriptions of Risk Rewards Themes [here](#)

BULL CASE	\$436.00	BASE CASE	\$331.00	BEAR CASE	\$219.00
~39x CY27 non-GAAP EPS \$11.18		~34x CY27 non-GAAP EPS of \$9.73		~25x CY27 non-GAAP EPS of \$8.76	
DRAM and NAND pricing and utilization improve further than expected, and LAM sees record revenue in CY27. The company sees revenue growth exceed WFE on the back of market share gains in leading edge nodes.		We expect LAM to outperform in 2026-27 on the back of NAND and foundry logic strength.		Memory markets turn weaker and LAM loses share to competitors.	
- Revenue growth of 45.6% in CY27 as LAM benefits from NAND WFE growth and market share gains		- Revenue growth of 31.2% in CY27 driven by NAND and foundry logic		- Revenue growth of 21.4% in CY27 as memory recovery more muted than expected	
- GM expands to 52.4% in CY27		- CY27 GM of 51.4% as regional mix deterioration is offset by top-line growth		- GM of 50.6% in CY27	

Risk Reward – Lam Research Corp (LRCX.O)

KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
GAAP Revenue (\$, mm)	18,436	23,173	31,494	38,491
MW Gross Margin (%)	48.8	50.2	51.0	51.6
Non-GAAP EPS (\$)	4.13	5.72	8.39	10.74
Inventory (\$, mm)	4,308	4,548	5,685	6,153
DOI	164.0	141.6	132.6	118.8

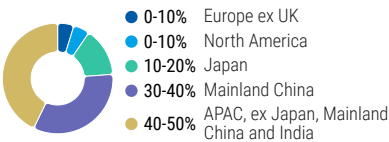
CATALYST CALENDAR

Date	Event	Source: Refinitiv, Morgan Stanley
04 Nov 2026 - 08 Nov 2026	Lam Research Corp Annual Shareholders Meeting	

INVESTMENT DRIVERS

- NAND WFE market trends (LAM has outsized share)
- Market share gains from new logic architectures (GAA, backside power)

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

5/5 BEST	24 Month Horizon	2/5 MOST	3 Month Horizon
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Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Market share gains in new logic architectures (GAA, backside power)
- NAND capex recovers to new highs
- Continued share gains in China

RISKS TO DOWNSIDE

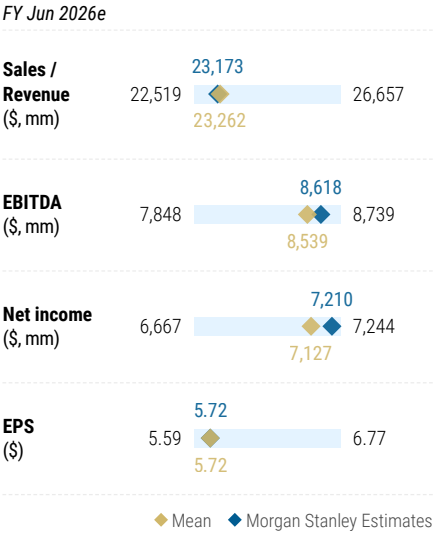
- NAND WFE weaker than expected
- Widespread export restrictions to China
- Market share loss to competitors in etch

OWNERSHIP POSITIONING

Inst. Owners, % Active	49.8%	
HF Sector Long/Short Ratio	2x	
HF Sector Net Exposure	25.3%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure – Short Exposure).

MS ESTIMATES VS. CONSENSUS



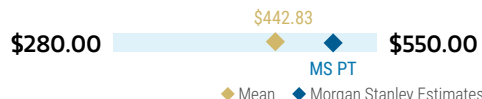
Risk Reward – Applied Materials Inc. (AMAT.O)

DRAM/Leading Logic upside & China/ICAPS derisked

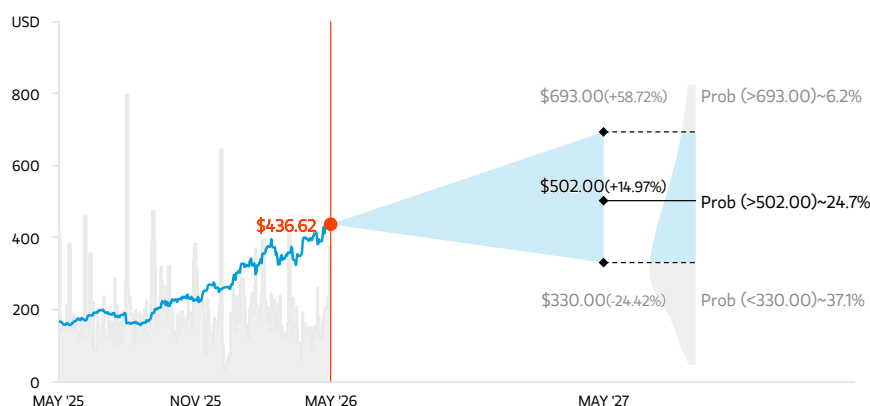
PRICE TARGET \$502.00

~28x CY27e EPS of \$17.94, a 6-turn discount to LAM and 5-turns to KLA to reflect growth prospects in DRAM but concerns around market share loss in China.

Consensus Price Target Distribution



RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)



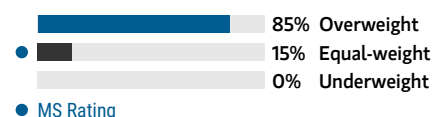
Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 15 May 2026. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology [here](#)

EQUAL-WEIGHT THESIS

■ AMAT has traded at a material valuation discount vs LAM & KLA given execution and concerns around market share loss in China. We think AMAT will be a share gainer in 2026 but given our growth forecast is in-line with broader WFE in 2027 we think the valuation discount is unlikely to narrow in the near term.

■ AMAT has the highest DRAM mix in their portfolio (CY26 31% vs LAM 23%, KLA 28%) and is most levered to greenfield wafer additions. We see DRAM as the fastest growing end-market in 2026 but the slowest growing in 2027.

Consensus Rating Distribution



Risk Reward Themes

New Data Era: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

\$693.00

~33x CY27 EPS of \$20.37

DRAM & ICAPS remain strong and AMAT's margin expansion accelerates.

- Revenue grows 36.9% in CY27 as AMAT outperforms WFE
- Gross margin expands to 51.3% in CY27
- Multiples expand to 33x PE, as equipment is perceived as a growth sector within semis

BASE CASE

\$502.00

~28x CY27 EPS of \$17.94

AMAT grows in-line with WFE with DRAM growth slightly offset by China/ICAPS

- Revenue growth of 24.5% in CY27
- Non-GAAP gross margin slightly increases to 50.8% in CY27
- Multiple at a premium to AMAT's historical average

BEAR CASE

\$330.00

~21x CY27 EPS of \$15.72

DRAM WFE remains weak in 2026 and leading edge logic sees headwinds from foundries scaling back their aspirations.

- Revenue increases 14.5% in CY27 as WFE recovery is subdued.
- Gross margin slightly increases to 50.1% in CY27
- Multiple contracts to a 21x PE, 1 standard deviation premium to historical levels

Risk Reward – Applied Materials Inc. (AMAT.O)

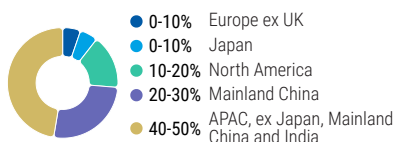
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
GAAP Revenue (\$, mm)	28,368	33,483	43,373	48,812
MW Gross Margin (%)	48.8	50.0	50.8	50.8
Non-GAAP EPS (\$)	9.42	12.31	17.02	19.71
Inventory (\$, mm)	5,915	7,580	9,137	9,786
DOI	146.2	162.8	154.0	146.5

INVESTMENT DRIVERS

- DRAM WFE outpaces overall WFE growth
- AMAT capitalizes on logic growth opportunities (GAA, back-side power)

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

5/5 BEST	24 Month Horizon	3/5 MOST	3 Month Horizon
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Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Market share gains as DRAM outpaces WFE
- AMAT captures market share in logic architecture inflections (backside power)

RISKS TO DOWNSIDE

- Widespread restrictions on equipment shipments and services to China
- AMAT loses market share with key customers (Samsung, TSMC)

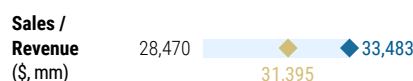
OWNERSHIP POSITIONING

Inst. Owners, % Active	51.7%	
HF Sector Long/Short Ratio	2x	
HF Sector Net Exposure	25.3%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure - Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Oct 2026e



◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Projected Financial Statements

Exhibit 38: LRCX: Projected Income Statement

Lam Research Corporation Income Statement (\$ in millions; fiscal year ends in June)	C3Q24A Sep-24	C4Q24E Dec-24	C1Q25A Mar-25	C2Q25A Jun-25	C3Q25A Sep-25	C4Q25A Dec-25	C1Q26A Mar-26	C2Q26E Jun-26	C3Q26E Sep-26	C4Q26E Dec-26	C1Q27E Mar-27	C2Q27E Jun-27	C3Q27E Sep-27	C4Q27E Dec-27	C1Q28E Mar-28	C2Q28E Jun-28	CY2024 Dec-24	CY2025 Dec-25	CY2026 Dec-26	CY2027 Dec-27	CY2028 Dec-28
Revenues - GAAP	\$ 4,168.0	\$ 4,376.0	\$ 4,720.2	\$ 5,171.4	\$ 5,324.2	\$ 5,344.8	\$ 5,841.5	\$ 6,662.2	\$ 7,082.9	\$ 7,480.8	\$ 8,107.5	\$ 8,822.9	\$ 9,219.0	\$ 9,363.4	\$ 9,884.9	\$ 10,023.6	\$ 16,209.1	\$ 20,560.5	\$ 27,067.4	\$ 35,512.7	\$ 40,216.9
q/q change (%)	7.7%	5.0%	7.9%	9.6%	3.0%	0.4%	9.3%	14.0%	6.3%	5.6%	8.4%	8.8%	4.5%	1.6%	5.6%	1.4%	13.2%	26.8%	31.6%	31.2%	13.2%
y/y change (%)	19.7%	16.4%	24.4%	33.6%	27.7%	22.1%	23.8%	28.8%	33.0%	40.0%	38.8%	32.4%	30.2%	25.2%	21.9%	13.6%					
COGS - GAAP	2,165.3	2,303.1	2,406.5	2,581.7	2,639.3	2,693.6	2,931.0	3,300.4	3,494.4	3,683.1	3,964.5	4,296.7	4,471.2	4,541.2	4,784.3	4,851.4	8,492.8	10,321.1	13,408.9	17,273.7	19,424.3
% of revenue	52.0%	52.6%	51.0%	50.1%	49.6%	50.4%	50.2%	49.5%	49.3%	49.2%	48.9%	48.7%	48.5%	48.5%	48.4%	48.4%	52.4%	50.2%	49.5%	48.6%	48.3%
COGS - non-GAAP	2,159.0	2,298.9	2,407.8	2,570.4	2,630.6	2,684.9	2,928.4	3,297.8	3,491.9	3,680.5	3,964.5	4,296.7	4,471.2	4,541.2	4,784.3	4,851.4	8,398.8	10,293.7	13,398.6	17,273.7	19,424.3
% of revenue	51.8%	52.5%	51.0%	49.7%	49.4%	50.2%	50.1%	49.5%	49.3%	49.2%	48.9%	48.7%	48.5%	48.5%	48.4%	48.4%	51.8%	50.1%	49.5%	48.6%	48.3%
Gross profit - GAAP	2,002.7	2,073.0	2,313.7	2,589.7	2,684.9	2,651.2	2,910.5	3,361.8	3,588.4	3,797.6	4,142.9	4,526.1	4,747.8	4,822.1	5,100.6	5,172.2	7,716.3	10,239.4	13,658.4	18,238.9	20,792.5
Gross margin (%)	48.0%	47.4%	49.0%	50.1%	50.4%	49.6%	49.8%	50.5%	50.7%	50.8%	51.1%	51.3%	51.5%	51.5%	51.6%	51.6%	47.6%	49.8%	50.5%	51.4%	51.7%
Gross profit - non-GAAP	2,009.0	2,077.2	2,312.4	2,601.0	2,693.6	2,659.9	2,913.1	3,364.4	3,591.0	3,800.2	4,142.9	4,526.1	4,747.8	4,822.1	5,100.6	5,172.2	7,810.3	10,266.8	13,668.8	18,238.9	20,792.5
Gross margin (%)	48.2%	47.5%	49.0%	50.3%	50.6%	49.8%	49.9%	50.5%	50.7%	50.8%	51.1%	51.3%	51.5%	51.5%	51.6%	51.6%	48.2%	49.9%	50.5%	51.4%	51.7%
R&D - GAAP	495.4	494.9	525.9	580.2	576.4	573.3	583.2	634.4	667.3	704.8	747.6	778.3	808.7	821.3	847.3	859.2	2,000.4	2,255.8	2,589.7	3,156.0	3,437.1
R&D - non-GAAP	487.2	492.5	533.1	564.8	565.6	565.3	585.4	634.4	667.3	704.8	747.6	778.3	808.7	821.3	847.3	859.2	1,959.9	2,228.8	2,591.9	3,156.0	3,437.1
% of revenue	11.7%	11.3%	11.3%	10.9%	10.6%	10.6%	10.0%	9.5%	9.4%	9.4%	9.2%	8.8%	8.8%	8.8%	8.6%	8.6%	12.1%	10.8%	9.6%	8.9%	8.5%
SG&A - GAAP	243.1	244.2	226.0	268.4	279.3	267.7	280.3	286.9	297.9	307.2	316.7	335.8	341.7	347.0	366.3	361.5	919.7	1,041.4	1,172.3	1,341.2	1,450.0
SG&A - non-GAAP	234.9	242.0	230.3	257.6	266.3	262.1	280.7	287.3	298.3	307.6	317.1	336.2	342.1	347.5	366.8	361.9	883.5	1,016.3	1,174.0	1,342.9	1,451.7
% of revenue	5.6%	5.5%	4.9%	5.0%	5.0%	4.9%	4.8%	4.3%	4.2%	4.1%	3.9%	3.8%	3.7%	3.7%	3.7%	3.6%	5.5%	4.9%	4.3%	3.8%	3.6%
Operating expenses - GAAP	738.5	739.1	751.9	848.6	855.8	841.0	863.5	921.2	965.2	1,012.0	1,064.3	1,114.1	1,150.3	1,168.3	1,213.6	1,220.7	2,934.5	3,297.3	3,762.0	4,497.1	4,887.1
Operating expenses - non-GAAP	722.1	734.5	763.3	822.4	831.9	827.5	866.2	921.7	965.7	1,012.4	1,064.8	1,114.6	1,150.8	1,168.8	1,214.1	1,221.1	2,843.5	3,245.1	3,765.9	4,498.9	4,888.9
q/q change (%)	4.8%	1.7%	3.9%	7.7%	1.2%	-0.5%	4.7%	6.4%	4.8%	4.8%	5.2%	4.7%	3.2%	1.6%	3.9%	0.6%	14.6%	14.1%	16.0%	19.5%	8.7%
% of revenue	17.3%	16.8%	16.2%	15.9%	15.6%	15.5%	14.8%	13.8%	13.6%	13.5%	13.1%	12.6%	12.5%	12.5%	12.3%	12.2%	17.5%	15.8%	13.9%	12.7%	12.2%
Operating income - GAAP	1,264.2	1,333.9	1,561.8	1,741.1	1,829.1	1,810.2	2,047.0	2,440.6	2,623.2	2,785.7	3,078.6	3,412.0	3,597.4	3,653.8	3,887.0	3,951.5	4,781.8	6,942.2	9,896.4	13,741.8	15,905.4
Operating margin (%)	30.3%	30.5%	33.1%	33.7%	34.4%	33.9%	35.0%	36.6%	37.0%	37.2%	38.0%	38.7%	39.0%	39.0%	39.3%	39.4%	29.5%	33.8%	36.6%	38.7%	39.5%
Operating income - non-GAAP	1,286.9	1,342.6	1,549.1	1,778.6	1,861.7	1,832.4	2,047.0	2,442.7	2,625.3	2,787.8	3,078.1	3,411.6	3,597.0	3,653.3	3,886.5	3,951.1	4,966.8	7,021.7	9,902.8	13,740.1	15,903.7
Operating margin (%)	30.9%	30.7%	32.8%	34.4%	35.0%	34.3%	35.0%	36.7%	37.1%	37.3%	38.0%	38.7%	39.0%	39.0%	39.3%	39.4%	30.6%	34.2%	36.6%	38.7%	39.5%
Total interest income (net) and other income - GAAP	30.1	14.3	(25.0)	37.9	30.1	26.4	(35.5)	25.0	25.0	25.0	25.0	25.0	25.0	25.0	25.0	25.0	108.2	69.3	39.5	100.0	100.0
Interest income (expense)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other income (expense), net	30.1	14.3	(25.0)	37.9	30.1	26.4	(35.5)	25.0	25.0	25.0	25.0	25.0	25.0	25.0	25.0	25.0	108.2	69.3	39.5	100.0	100.0
Total interest income (net) and other income - non-GAAP	13.4	10.5	(7.4)	4.4	14.1	10.5	(7.5)	52.9	52.9	52.9	52.9	52.9	52.9	52.9	52.9	52.9	53.2	21.7	151.3	211.8	211.8
PBT - GAAP	1,294.3	1,348.1	1,536.7	1,779.0	1,859.2	1,836.6	2,011.6	2,465.6	2,648.2	2,810.7	3,103.6	3,437.0	3,622.4	3,678.8	3,912.0	3,976.5	4,890.0	7,011.5	9,936.0	13,841.8	16,005.4
PBT - non-GAAP	1,300.3	1,353.2	1,541.7	1,783.0	1,875.8	1,842.9	2,039.4	2,495.7	2,678.3	2,840.8	3,131.1	3,464.5	3,649.9	3,706.3	3,939.5	4,004.0	5,020.0	7,043.4	10,054.1	13,951.8	16,115.4
Income tax expense (benefit) - GAAP	177.8	157.1	206.1	58.9	290.5	242.6	186.1	345.2	370.7	393.5	434.5	481.2	507.1	515.0	547.7	556.7	596.4	798.1	1,295.5	1,937.9	2,240.8
Income tax expense (benefit) - non-GAAP	178.8	178.2	205.7	85.3	264.8	243.6	188.0	347.1	372.6	395.4	436.4	483.1	509.0	516.9	549.6	558.6	632.4	799.4	1,303.1	1,945.4	2,248.3
Effective tax rate (%)	13.8%	13.2%	13.3%	4.8%	14.1%	13.2%	9.2%	13.9%	13.9%	13.9%	13.9%	13.9%	13.9%	13.9%	14.0%	14.0%	12.6%	11.3%	13.0%	13.9%	14.0%
Net income, cont. ops. - GAAP	1,116.4	1,191.0	1,330.7	1,720.1	1,568.7	1,594.0	1,825.5	2,120.4	2,277.4	2,417.2	2,669.1	2,955.8	3,115.3	3,163.8	3,364.3	3,419.8	4,293.6	6,213.4	8,640.5	11,904.0	13,764.6
Net income, cont. ops. - non-GAAP	1,121.5	1,175.0	1,336.0	1,697.7	1,611.1	1,699.2	1,851.4	2,148.6	2,305.6	2,445.4	2,694.7	2,981.4	3,140.9	3,189.4	3,389.9	3,445.4	4,387.6	6,244.0	8,751.0	12,006.4	13,867.1
Net margin (%)	26.9%	26.9%	28.3%	32.8%	30.3%	29.9%	31.7%	32.3%	32.6%	32.7%	33.2%	33.8%	34.1%	34.1%	34.3%	34.4%	27.1%	30.4%	32.3%	33.8%	34.5%
EPS - GAAP	\$ 0.86	\$ 0.92	\$ 1.03	\$ 1.35	\$ 1.24	\$ 1.26	\$ 1.45	\$ 1.69	\$ 1.82	\$ 1.94	\$ 2.15	\$ 2.39	\$ 2.53	\$ 2.58	\$ 2.75	\$ 2.80	\$ 3.29	\$ 4.88	\$ 6.91	\$ 9.65	\$ 11.29
EPS - non-GAAP	\$ 0.86	\$ 0.91	\$ 1.04	\$ 1.33	\$ 1.27	\$ 1.27	\$ 1.47	\$ 1.71	\$ 1.85	\$ 1.97	\$ 2.17	\$ 2.41	\$ 2.55	\$ 2.60	\$ 2.77	\$ 2.82	\$ 3.36	\$ 4.90	\$ 7.00	\$ 9.73	\$ 11.38

Source: Company data, Morgan Stanley Research estimates

Exhibit 39: LRCX: Projected Balance Sheet

Lam Research Corporation Balance Sheet (\$ in millions; fiscal year ends in June)	C3Q24A Sep-24 2025/1F	C4Q24E Dec-24 2025/2F	C1Q25A Mar-25 2025/3F	C2Q25A Jun-25 2025/4F	C3Q25A Sep-25 2026/1F	C4Q25A Dec-25 2026/2F	C1Q26A Mar-26 2026/3F	C2Q26E Jun-26 2026/4F	C3Q26E Sep-26 2027/1F	C4Q26E Dec-26 2027/2F	C1Q27E Mar-27 2027/3F	C2Q27E Jun-27 2027/4F	C3Q27E Sep-27 2028/1F	C4Q27E Dec-27 2028/2F	C1Q28E Mar-28 2028/3F	C2Q28E Jun-28 2028/4F	CY2024 Dec-24	CY2025 Dec-25	CY2026 Dec-26	CY2027 Dec-27	CY2028 Dec-28
Assets:																					
Cash and cash equivalents	6,067.5	5,665.4	5,450.7	6,390.7	6,693.0	6,180.4	4,750.9	4,568.1	4,477.2	4,482.4	4,289.4	4,834.1	5,239.1	6,190.3	6,243.2	8,189.7	5,665.4	6,180.4	4,482.4	6,190.3	10,449.4
Marketable securities	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Accounts receivable, net	2,937.2	3,304.9	3,228.2	3,378.1	3,633.0	3,492.0	4,132.9	4,467.7	4,576.5	4,975.8	5,606.8	5,783.1	5,988.8	5,967.5	6,935.2	6,004.8	3,304.9	3,492.0	4,975.8	5,967.5	6,118.1
Inventory	4,209.9	4,358.2	4,463.3	4,308.0	4,095.1	4,037.7	4,000.0	4,548.2	4,763.3	5,020.5	5,303.9	5,685.2	5,851.7	5,943.3	6,134.8	6,152.5	4,358.2	4,037.7	5,020.5	5,943.3	6,166.3
Deferred income taxes	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other current assets	277.8	284.4	318.1	440.3	385.6	307.9	413.1	413.1	413.1	413.1	413.1	413.1	413.1	413.1	413.1	413.1	284.4	307.9	413.1	413.1	413.1
Total current assets	13,492.4	13,612.8	13,460.3	14,517.0	14,806.7	14,018.0	13,296.9	13,997.2	14,230.1	14,891.8	15,613.3	16,715.5	17,492.7	18,514.2	19,726.3	20,760.1	13,612.8	14,018.0	14,891.8	18,514.2	23,146.9
PP&E, net	2,214.3	2,313.6	2,372.2	2,428.7	2,510.5	2,711.0	2,853.6	3,041.2	3,241.4	3,452.7	3,685.3	3,942.3	4,208.9	4,473.2	4,752.2	5,028.0	2,313.6	2,711.0	3,452.7	4,473.2	5,563.0
Restricted cash	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Long-term investments	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Deferred income taxes	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Goodwill	1,619.8	1,622.5	1,656.7	1,670.1	1,688.4	1,725.5	1,743.5	1,743.5	1,743.5	1,743.5	1,743.5	1,743.5	1,743.5	1,743.5	1,743.5	1,743.5	1,622.5	1,725.5	1,743.5	1,743.5	1,743.5
Intangible assets, net	138.5	138.5	138.5	138.5	138.5	138.5	138.5	138.5	138.5	138.5	138.5	138.5	138.5	138.5	138.5	138.5	138.5	138.5	138.5	138.5	138.5
Other non-current assets	2,067.5	2,152.5	2,340.5	2,590.8	2,756.0	2,798.1	2,759.4	2,759.4	2,759.4	2,759.4	2,759.4	2,759.4	2,759.4	2,759.4	2,759.4	2,759.4	2,152.5	2,798.1	2,759.4	2,759.4	2,759.4
Total assets	19,532.5	19,839.9	19,968.3	21,345.3	21,900.2	21,391.2	20,791.9	21,679.7	22,112.8	22,985.9	23,939.9	25,299.2	26,343.0	27,628.8	29,119.8	30,429.5	19,839.9	21,391.2	22,985.9	27,628.8	33,351.3
Liabilities & Shareholders' Equity:																					
Accounts payable	704.2	822.3	853.3	854.2	863.2	1,026.9	1,071.6	1,265.5	1,182.1	1,397.6	1,440.9	1,601.7	1,530.1	1,650.7	1,774.8	1,660.6	822.3	1,026.9	1,397.6	1,650.7	1,690.6
Accrued expenses and other current liabilities	2,196.4	2,096.0	1,999.9	2,394.4	2,443.9	2,267.5	2,075.4	2,075.4	2,075.4	2,075.4	2,075.4	2,075.4	2,075.4	2,075.4	2,075.4	2,075.4	2,096.0	2,267.5	2,075.4	2,075.4	2,075.4
Deferred profit	1,937.3	1,927.9	1,882.3	2,565.5	2,646.4	2,164.7	2,091.3	2,091.3	2,091.3	2,091.3	2,091.3	2,091.3	2,091.3	2,091.3	2,091.3	2,091.3	1,927.9	2,164.7	2,091.3	2,091.3	2,091.3
Income taxes payable	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other liabilities	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Current portion of long-term debt	504.7	504.1	754.3	754.3	754.4	754.0	4.1	4.1	4.1	4.1	4.1	4.1	4.1	4.1	4.1	4.1	504.1	754.0	4.1	4.1	4.1
Total current liabilities	5,342.7	5,350.3	5,489.8	6,568.4	6,707.9	6,213.2	5,242.4	5,436.3	5,352.9	5,568.4	5,611.7	5,772.5	5,700.8	5,821.5	5,945.6	5,831.4	5,350.3	6,213.2	5,568.4	5,821.5	5,861.4
Long-term debt	4,479.1	4,478.1	3,730.0	3,730.2	3,729.6	3,729.7	3,730.4	3,730.4	3,730.4	3,730.4	3,730.4	3,730.4	3,730.4	3,730.4	3,730.4	3,730.4	4,478.1	3,729.7	3,730.4	3,730.4	3,730.4
Deferred profit, non-current	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Income taxes payable	664.7	669.7	690.7	603.4	646.0	667.6	621.6	621.6	621.6	621.6	621.6	621.6	621.6	621.6	621.6	621.6	669.7	667.6	621.6	621.6	621.6
Other long-term liabilities	574.1	533.7	546.7	581.6	623.9	635.2	612.8	612.8	612.8	612.8	612.8	612.8	612.8	612.8	612.8	612.8	533.7	635.2	612.8	612.8	612.8
Minority interest	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total liabilities	11,060.6	11,031.9	10,457.2	11,483.6	11,707.5	11,245.7	10,207.1	10,401.0	10,317.6	10,533.2	10,576.5	10,737.2	10,665.6	10,786.3	10,910.3	10,796.2	11,031.9	11,245.7	10,533.2	10,786.3	10,826.1
Preferred stock	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Common stock	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3
Additional paid-in capital	8,303.0	8,439.9	8,529.0	8,697.3	8,794.5	8,948.4	9,054.2	9,150.8	9,260.0	9,369.2	9,478.4	9,587.5	9,707.6	9,827.7	9,947.8	10,067.9	8,439.9	8,948.4	9,369.2	9,827.7	10,332.1
Total share capital	8,304.3	8,441.2	8,530.3	8,698.6	8,795.8	8,949.7	9,055.5	9,152.1	9,261.3	9,370.4	9,479.6	9,588.8	9,708.9	9,829.0	9,949.1	10,069.2	8,441.2	8,949.7	9,370.4	9,829.0	10,333.4
Treasury stock	(25,374.7)	(26,024.1)	(26,455.9)	(27,763.4)	(28,759.5)	(30,203.8)	(31,347.7)	(32,547.7)	(34,047.7)	(35,547.7)	(37,047.7)	(38,547.7)	(40,247.7)	(41,947.7)	(43,647.7)	(45,347.7)	(26,024.1)	(30,203.8)	(35,547.7)	(41,947.7)	(48,747.7)
Temporary equity, convertible notes	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Retained earnings (accumulated deficit)	25,630.0	26,525.0	27,560.6	28,988.9	30,230.1	31,498.2	32,998.4	34,795.7	36,703.0	38,751.4	41,052.9	43,642.3	46,337.6	49,082.7	52,029.5	55,033.2	26,525.0	31,498.2	38,751.4	49,082.7	61,060.9
Accumulated other comprehensive income (loss)	(87.8)	(134.1)	(123.9)	(62.4)	(73.6)	(98.7)	(121.4)	(121.4)	(121.4)	(121.4)	(121.4)	(121.4)	(121.4)	(121.4)	(121.4)	(121.4)	(134.1)	(98.7)	(121.4)	(121.4)	(121.4)
Plug for earnings	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total shareholders' equity	8,471.9	8,808.0	9,511.1	9,861.6	10,192.8	10,145.4	10,584.8	11,278.7	11,795.2	12,452.7	13,363.5	14,562.0	15,677.4	16,842.6	18,209.5	19,633.3	8,808.0	10,145.4	12,452.7	16,842.6	22,525.2
Total liabilities & shareholders' equity	19,532.5	19,839.9	19,968.3	21,345.3	21,900.2	21,391.2	20,791.9	21,679.7	22,112.8	22,985.9	23,939.9	25,299.2	26,343.0	27,628.8	29,119.8	30,429.5	19,839.9	21,391.2	22,985.9	27,628.8	33,351.3

Source: Company data, Morgan Stanley Research estimates

Exhibit 40: LRCX: Projected Cash Flow Statement

Lam Research Corporation Cash Flow Statement (\$ in millions, fiscal year ends in June)	C3Q24A Sep-24 2025/1F	C4Q24E Dec-24 2025/2F	C1Q25A Mar-25 2025/3F	C2Q25A Jun-25 2025/4F	C3Q25A Sep-25 2026/1F	C4Q25A Dec-25 2026/2F	C1Q26A Mar-26 2026/3F	C2Q26E Jun-26 2026/4F	C3Q26E Sep-26 2027/1F	C4Q26E Dec-26 2027/2F	C1Q27E Mar-27 2027/3F	C2Q27E Jun-27 2027/4F	C3Q27E Sep-27 2028/1F	C4Q27E Dec-27 2028/2F	C1Q28E Mar-28 2028/3F	C2Q28E Jun-28 2028/4F	CY2024 Dec-24	CY2025 Dec-25	CY2026 Dec-26	CY2027 Dec-27	CY2028 Dec-28
Cash flows provided by (used for) operating activities:																					
Net income	\$ 1,116.4	\$ 1,191.0	\$ 1,330.7	\$ 1,720.1	\$ 1,568.7	\$ 1,594.0	\$ 1,825.5	\$ 2,120.4	\$ 2,277.4	\$ 2,417.2	\$ 2,669.1	\$ 2,955.8	\$ 3,115.3	\$ 3,163.8	\$ 3,364.3	\$ 3,419.8	\$ 4,293.6	\$ 6,213.4	\$ 8,640.5	\$ 11,904.0	\$ 13,764.6
<i>Adjustments to reconcile net income to net cash provided by operating activities:</i>																					
Depreciation and amortization	94.3	96.2	97.3	98.4	101.6	103.9	116.3	112.2	118.5	125.2	132.3	140.0	148.3	157.0	165.9	175.2	368.8	401.4	472.4	577.6	720.0
Stock-based compensation expense	80.0	82.0	87.1	94.3	97.2	88.5	96.6	96.6	109.2	109.2	109.2	109.2	120.1	120.1	120.1	120.1	317.9	367.2	411.6	458.5	504.4
Tax effect from stock-based compensation	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Loss (gain) on equity investments, net	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Restructuring charges and asset impairments	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Gain/Loss on disposal of property, plant and equipment	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Deferred income taxes	(108.7)	(82.9)	(20.0)	(151.7)	(62.9)	(31.0)	(19.5)	-	-	-	-	-	-	-	-	-	(277.6)	(265.5)	(19.5)	-	-
Other non-cash items	(0.5)	(8.6)	1.7	14.2	(1.9)	(20.0)	(2.9)	-	-	-	-	-	-	-	-	-	(2.8)	(6.0)	(2.9)	-	-
<i>Changes in current assets and liabilities:</i>																					
Accounts receivable	-	-	-	(858.7)	-	-	-	(334.8)	(108.8)	(399.3)	(631.0)	(176.3)	(205.6)	21.3	(967.7)	930.4	303.4	(858.7)	(842.9)	(991.7)	(150.7)
Inventory	-	-	-	(180.7)	-	-	-	(548.2)	(215.1)	(257.2)	(283.4)	(381.3)	(166.5)	(91.6)	(191.5)	(17.7)	528.7	(180.7)	(1,020.5)	(922.8)	(222.9)
Income taxes receivable	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other current assets	-	-	-	(206.7)	-	-	-	-	-	-	-	-	-	-	-	-	(15.5)	(206.7)	-	-	-
Accounts payable	-	-	-	212.0	-	-	-	193.9	(83.4)	215.5	43.3	160.8	(71.6)	120.7	124.0	(114.1)	125.9	212.0	326.0	253.1	39.9
Accrued expenses and other current liabilities	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Income taxes payable	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Deferred revenues	-	-	-	1,147.8	-	-	-	-	-	-	-	-	-	-	-	-	(277.4)	1,147.8	-	-	-
Other changes in assets and liabilities	386.9	(535.8)	(188.1)	665.3	76.2	(255.5)	(874.6)	-	-	-	-	-	-	-	-	-	(807.3)	297.8	(874.6)	-	-
Net cash provided by operating activities	1,568.5	741.9	1,308.7	2,554.2	1,779.0	1,480.0	1,141.4	1,640.1	2,097.9	2,210.6	2,039.4	2,808.1	2,939.9	3,491.2	2,615.1	4,513.7	4,557.7	7,121.9	7,090.0	11,278.7	14,655.3
Cash flows from investing activities:																					
Capital expenditures, net	(110.6)	(188.3)	(288.1)	(172.2)	(185.1)	(260.9)	(331.6)	(299.8)	(318.7)	(336.6)	(364.8)	(397.0)	(414.9)	(421.4)	(444.8)	(451.1)	(503.3)	(906.2)	(1,286.8)	(1,598.1)	(1,809.8)
Transfer of restricted cash and investments	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Purchase (sale) of available-for-sale securities, net	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	14.7	-	-	-	-
Purchase of long-term investments	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Purchase of intangible assets	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Proceeds from sale of business, net	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other investing activities	0.0	13.0	(4.9)	42.9	(0.9)	3.1	(3.0)	-	-	-	-	-	-	-	-	-	8.8	40.3	(3.0)	-	-
Net cash used for investing activities	(110.6)	(175.4)	(292.9)	(129.3)	(186.0)	(257.8)	(334.6)	(299.8)	(318.7)	(336.6)	(364.8)	(397.0)	(414.9)	(421.4)	(444.8)	(451.1)	(479.9)	(866.0)	(1,289.7)	(1,598.1)	(1,809.8)
Cash flows from financing activities:																					
Proceeds from (payments of) long-term debt	(0.9)	(1.0)	(504.0)	(1.5)	(1.4)	(1.5)	(751.2)	-	-	-	-	-	-	-	-	-	(4.0)	(508.4)	(751.2)	-	-
Proceeds from common stock issuances	(0.0)	(0.2)	2.0	0.7	-	-	13.0	-	-	-	-	-	-	-	-	-	10.8	2.7	13.0	-	-
Proceeds from (payments of) short-term debt	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Payment of dividends to stockholders	(261.0)	(297.6)	(295.7)	(295.2)	(292.0)	(327.5)	(325.8)	(323.1)	(370.1)	(368.8)	(367.5)	(366.4)	(420.0)	(418.7)	(417.4)	(416.1)	(1,082.8)	(1,210.4)	(1,387.8)	(1,572.7)	(1,786.4)
Treasury stock repurchases	(997.0)	(697.7)	(435.3)	(1,292.3)	(975.8)	(1,466.2)	(1,162.8)	(1,200.0)	(1,500.0)	(1,500.0)	(1,500.0)	(1,500.0)	(1,700.0)	(1,700.0)	(1,700.0)	(1,700.0)	(3,048.8)	(4,169.5)	(5,362.8)	(6,400.0)	(6,800.0)
Other financing activities	(0.3)	61.3	(1.4)	80.7	(12.4)	69.9	(3.8)	-	-	-	-	-	-	-	-	-	120.3	136.7	(3.8)	-	-
Net cash used for financing activities	(1,259.3)	(935.2)	(1,234.5)	(1,507.6)	(1,281.6)	(1,725.2)	(2,230.6)	(1,523.1)	(1,870.1)	(1,868.8)	(1,867.5)	(1,866.4)	(2,120.0)	(2,118.7)	(2,117.4)	(2,116.1)	(4,004.5)	(5,748.9)	(7,492.6)	(7,972.7)	(8,586.4)
Effect of exchange rate changes on cash and cash equivalents	22.7	(26.0)	2.4	29.3	(7.1)	(13.3)	(5.0)	-	-	-	-	-	-	-	-	-	(21.4)	11.3	(5.0)	-	-
Cash and cash equivalents, beginning of period	5,847.9	6,067.5	5,665.4	5,450.7	6,390.7	6,693.0	6,180.4	4,750.9	4,568.1	4,477.2	4,482.4	4,289.4	4,834.1	5,239.1	6,190.3	6,243.2	5,623.3	5,665.4	6,180.4	4,482.4	6,190.3
Change in cash flow due to restatement	(1.7)	(7.4)	1.7	(6.7)	(1.8)	3.7	(0.7)	-	-	-	-	-	-	-	-	-	(9.8)	(3.1)	(0.7)	-	-
Net increase (decrease) in cash and cash equivalents	221.3	(394.7)	(216.4)	946.6	304.2	(516.3)	(1,428.8)	(182.8)	(91.0)	5.2	(192.9)	544.7	405.0	951.2	52.9	1,946.5	51.9	518.2	(1,697.3)	1,707.9	4,259.1
Cash and cash equivalents, end of period	6,067.5	5,665.4	5,450.7	6,390.7	6,693.0	6,180.4	4,750.9	4,568.1	4,477.2	4,482.4	4,289.4	4,834.1	5,239.1	6,190.3	6,243.2	8,189.7	5,665.4	6,180.4	4,482.4	6,190.3	10,449.4

Source: Company data, Morgan Stanley Research estimates

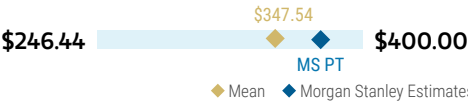
Risk Reward – MKS Inc. (MKSIO) Top Pick

Cyclical recovery to drive deleverage & EPS growth, OW

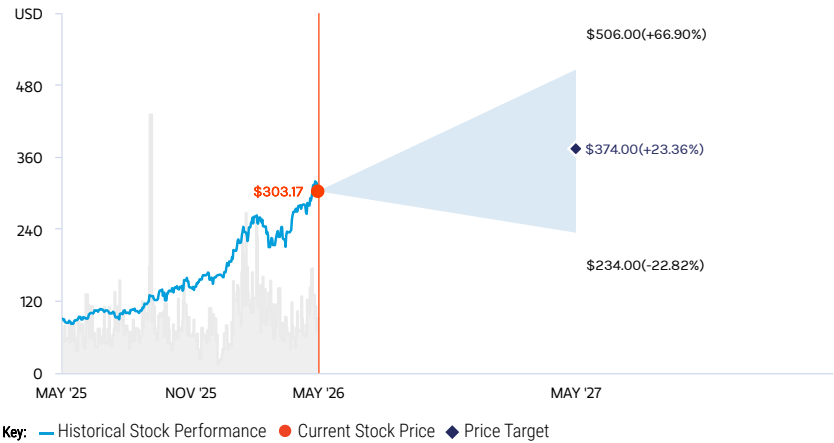
PRICE TARGET **\$374.00**

22x FY27 EPS of \$17.01. Multiple is a larger discount to front-end SPE vs historic to reflect balance sheet concerns and business mix shift to packaging.

Consensus Price Target Distribution



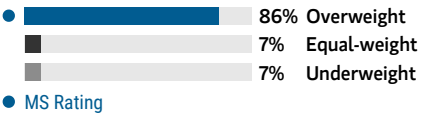
RISK REWARD CHART



OVERWEIGHT THESIS

- Cyclical recoveries for MKS' semi and electronics & packaging (E&P) business should drive revenue and FCF growth, helping the company delever and EPS outperform.
- The company's outsized NAND exposure has worked against the company during 2023-2024, but we expect NAND spend to outgrow overall WFE in 2026-2027 and MKS to be a beneficiary.
- We forecast MKS' E&P business to grow above PCBs but below advanced electronics through 2027, as the company's "Optimize the Interconnect" strategy drives share gains from scale.

Consensus Rating Distribution



Risk Reward Themes

New Data Era: *Positive*
Self-help: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE **\$506.00**

25x FY27 EPS of \$20.24

A stronger than expected WFE and advanced electronics upcycle drives more aggressive debt paydowns and MKS' EPS sees strong growth.

-Revenue grows 29.9% in 2027 on the back of a stronger WFE and advanced electronics upcycle.

-GPM% expands to 48.6% and EPS to \$20.24 in 2027.

BASE CASE **\$374.00**

22x FY27 EPS of \$17.01

MKS' semiconductor & E&P businesses benefit from a cyclical recovery in 2027 and grows revenue y/y in both segments.

-Revenue grows 21.4% in 2027.

-GPM% expands to 47.6% and EPS to \$17.01 in 2027.

BEAR CASE **\$234.00**

16x FY27 EPS of \$14.62

A weaker WFE and advanced electronics cycle dampen the recovery and MKS sees slower debt repayments and deleverage.

-Revenue growth of 12.9% in 2027 on the back of a weaker WFE and advanced electronics upcycle.

-GPM% grows to 46.1% and EPS to \$14.62 in 2027.

Risk Reward – MKS Inc. (MKSIO)

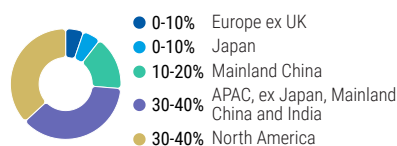
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
GAAP Revenue (\$, mm)	3,930	4,793	5,821	6,322
MW Gross Margin (%)	46.8	47.0	47.6	47.8
MW EPS (\$)	7.88	11.70	17.01	19.18
Inventory (\$, mm)	651	914	1,096	1,150
DOI (\$, mm)	158.5	153.4	150.9	145.8

INVESTMENT DRIVERS

- WFE and semiconductor subcomponent cycle.
- MKS' market share in semi subcomponents (outsized exposure to NAND).
- Advanced Electronics and PCB & Substrate cycle.

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

5/5 BEST	24 Month Horizon	4/5 MOST	3 Month Horizon
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Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Stronger than expected WFE spend
- Market share gains in semis on the back of NAND growth
- PC, smartphone, and server strength
- Debt paydowns progress quicker than anticipated

RISKS TO DOWNSIDE

- WFE growth slows and NAND makers continue to be disciplined on capex
- PC, smartphone, and server extended weakness
- Debt paydowns slow due to slow revenue growth and higher rates

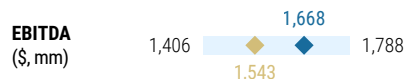
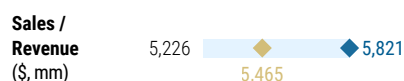
OWNERSHIP POSITIONING

Inst. Owners, % Active	65%	
HF Sector Long/Short Ratio	2x	
HF Sector Net Exposure	25.3%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure - Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Dec 2027e



◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Valuation Methodology and Risks

Intel Corporation (INTC.O)

~42x EPS CY2027 EPS of \$1.73, 42x is above the high end of the large cap logic semi peer group, reflecting high leverage potential on numbers that are still depressed, and foundry optionality, despite our longer term skepticism

Risks to Upside

- Foundry partnerships de-risk the story and further improve the multiple
- The company regains lost share in desktop and server following CPU shortages

Risks to Downside

- AMD competition increasingly becomes more significant, which could lead to further share losses in processors and pressure on ASPs
- Minimal success in foundry leads to an inflated cost structure

KLA Corp (KLAC.O)

33x CY27e EPS of \$57.58, a 15% premium to US SPE peers LAM & AMAT to reflect growth prospects in foundry logic and DRAM.

Risks to Upside

- Foundries (TSMC, Intel, Samsung) pull forward their logic roadmaps
- Design starts in leading edge nodes increase
- Process control intensity increases in DRAM due to HBM

Risks to Downside

- Certain foundries (Intel, Samsung) scale back their aspirations
- Market share loss in mask inspection to Lasertec or in e-beam to AMAT or ASML
- China restrictions

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - [Options_Probabilities_Exhibit_Link.pdf](#)
2. View descriptions of Risk Rewards Themes - [RR_Themes_Exhibit_Link.pdf](#)
3. View explanation of regional hierarchies - [GEG_Exhibit_Link.pdf](#)
4. View explanation of Theme/Exposure methodology - [ESG_Sustainable_Solutions_External_Link.pdf](#)
5. View explanation of HERS methodology - [ESG_HERS_External_Link.pdf](#)

Disclosure Section

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Global Stock Ratings Distribution

(as of April 30, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1546	42%	467	51%	30%	709	44%
Equal-weight/Hold	1568	43%	358	39%	23%	715	44%
Not-Rated/Hold	4	0%	0	0%	0%	1	0%
Underweight/Sell	555	15%	84	9%	15%	202	12%
Total	3,673		909			1627	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

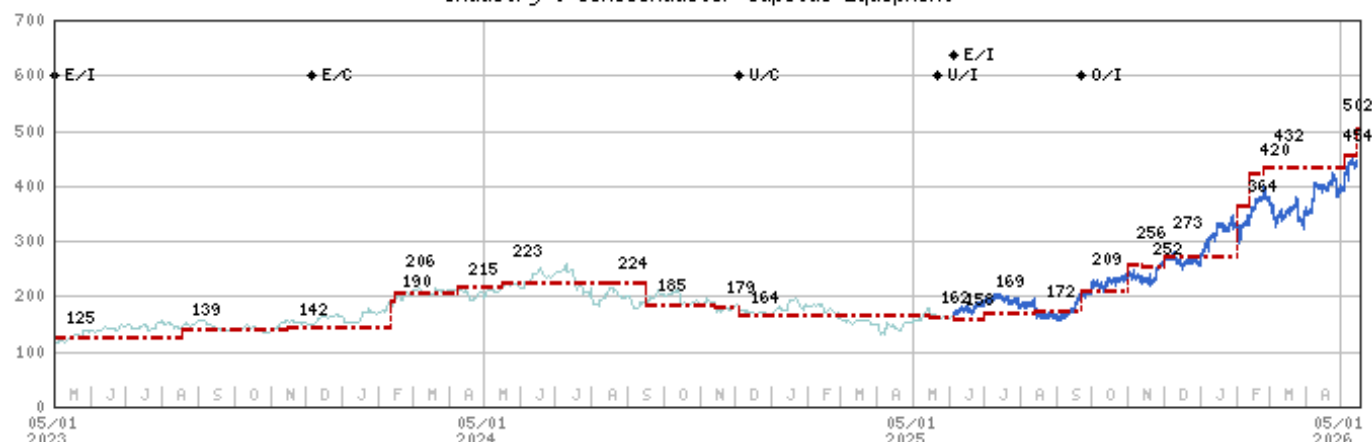
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Applied Materials Inc. (AMAT.O) - As of 05/17/26 GMT in USD
Industry : Semiconductor Capital Equipment



Stock Rating History: 5/1/21 : E/I; 12/7/23 : E/C; 12/5/24 : U/C; 5/22/25 : U/I; 6/5/25 : E/I; 9/22/25 : O/I

Price Target History: 2/19/21 : 113; 5/17/21 : 137; 5/21/21 : 139; 8/20/21 : 150; 12/14/21 : 160; 5/19/22 : 133; 7/24/22 : 115; 9/6/22 : 109; 10/13/22 : 99; 10/17/22 : 89; 11/18/22 : 86; 2/16/23 : 125; 8/17/23 : 139; 11/16/23 : 142; 2/12/24 : 190; 2/15/24 : 206; 4/8/24 : 215; 5/16/24 : 223; 8/15/24 : 224; 9/16/24 : 185; 11/14/24 : 179; 12/5/24 : 164; 5/15/25 : 162; 6/5/25 : 158; 7/1/25 : 169; 8/14/25 : 172; 9/22/25 : 209; 10/31/25 : 256; 11/13/25 : 252; 12/1/25 : 273; 2/2/26 : 364; 2/12/26 : 420; 2/24/26 : 432; 5/4/26 : 454; 5/14/26 : 502

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

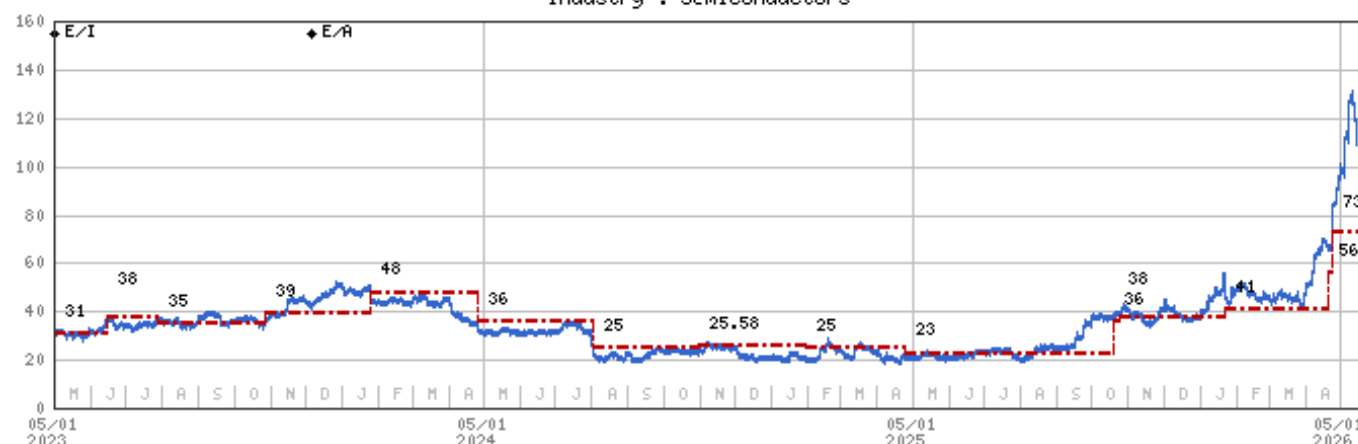
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Intel Corporation (INTC.O) - As of 05/17/26 GMT in USD
Industry : Semiconductors



Stock Rating History: 5/1/21 : O/I; 10/22/21 : E/I; 3/3/22 : U/I; 2/22/23 : E/I; 12/7/23 : E/A

Price Target History: 1/22/21 : 72; 7/23/21 : 70; 10/5/21 : 67; 10/22/21 : 55; 3/3/22 : 47; 4/28/22 : 46; 7/28/22 : 36; 10/27/22 : 29.5; 2/22/23 : 28; 4/27/23 : 31; 6/15/23 : 38; 7/28/23 : 35; 10/27/23 : 39; 1/26/24 : 48; 4/26/24 : 36; 8/2/24 : 25; 10/31/24 : 25.58; 1/31/25 : 25; 4/25/25 : 23; 10/20/25 : 36; 10/24/25 : 38; 1/23/26 : 41; 4/20/26 : 56; 4/24/26 : 73

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

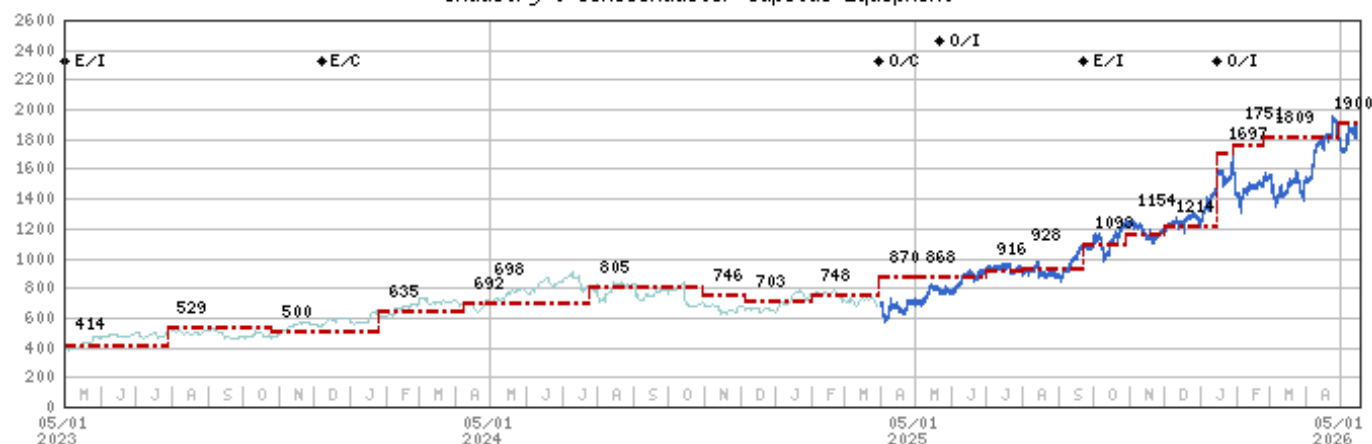
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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KLA Corp (KLAC.O) - As of 05/17/26 GMT in USD
Industry : Semiconductor Capital Equipment



Stock Rating History: 5/1/21 : E/I; 12/7/23 : E/C; 3/30/25 : O/C; 5/22/25 : O/I; 9/22/25 : E/I; 1/15/26 : O/I

Price Target History: 4/30/21 : 364; 7/30/21 : 389; 10/28/21 : 399; 12/14/21 : 431; 1/28/22 : 400; 4/25/22 : 375; 4/29/22 : 373; 7/24/22 : 377; 7/28/22 : 381; 9/6/22 : 358; 10/17/22 : 289; 10/27/22 : 318; 1/27/23 : 420; 4/26/23 : 414; 7/28/23 : 529; 10/26/23 : 500; 1/26/24 : 635; 4/8/24 : 692; 4/26/24 : 698; 7/24/24 : 805; 10/30/24 : 746; 12/5/24 : 703; 1/31/25 : 748; 3/30/25 : 870; 5/1/25 : 868; 7/1/25 : 916; 7/31/25 : 928; 9/22/25 : 1093; 10/29/25 : 1154; 12/1/25 : 1214; 1/15/26 : 1697; 1/29/26 : 1751; 2/24/26 : 1809; 4/29/26 : 1900

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

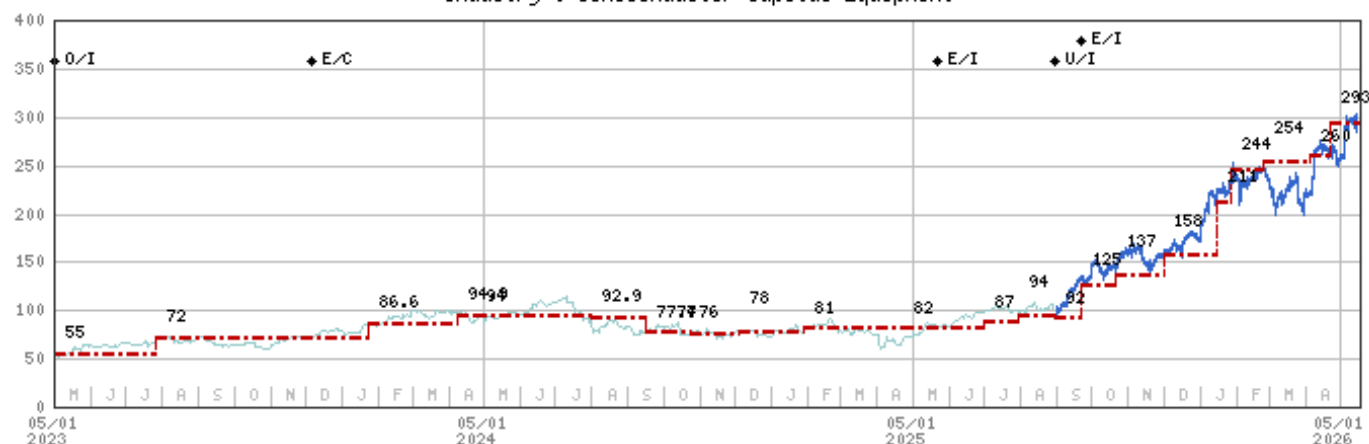
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Lam Research Corp (LRCX.O) - As of 05/17/26 GMT in USD
Industry : Semiconductor Capital Equipment



Stock Rating History: 5/1/21 : O/I; 12/1/23 : E/C; 5/22/25 : E/I; 8/31/25 : U/I; 9/22/25 : E/I

Price Target History: 4/22/21 : 72.6; 10/21/21 : 71; 12/14/21 : 75.5; 1/26/22 : 73; 4/20/22 : 62.5; 7/24/22 : 55; 9/6/22 : 54.3; 10/17/22 : 42.9; 10/19/22 : 40.4; 1/26/23 : 55; 7/26/23 : 72; 1/24/24 : 86.6; 4/8/24 : 94.9; 4/24/24 : 94; 7/31/24 : 92.9; 9/16/24 : 77.4; 10/3/24 : 77; 10/23/24 : 76; 12/6/24 : 78; 1/29/25 : 81; 4/23/25 : 82; 7/1/25 : 87; 7/30/25 : 94; 8/31/25 : 92; 9/22/25 : 125; 10/22/25 : 137; 12/1/25 : 158; 1/15/26 : 211; 1/28/26 : 244; 2/24/26 : 254; 4/5/26 : 260; 4/22/26 : 293

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target — No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

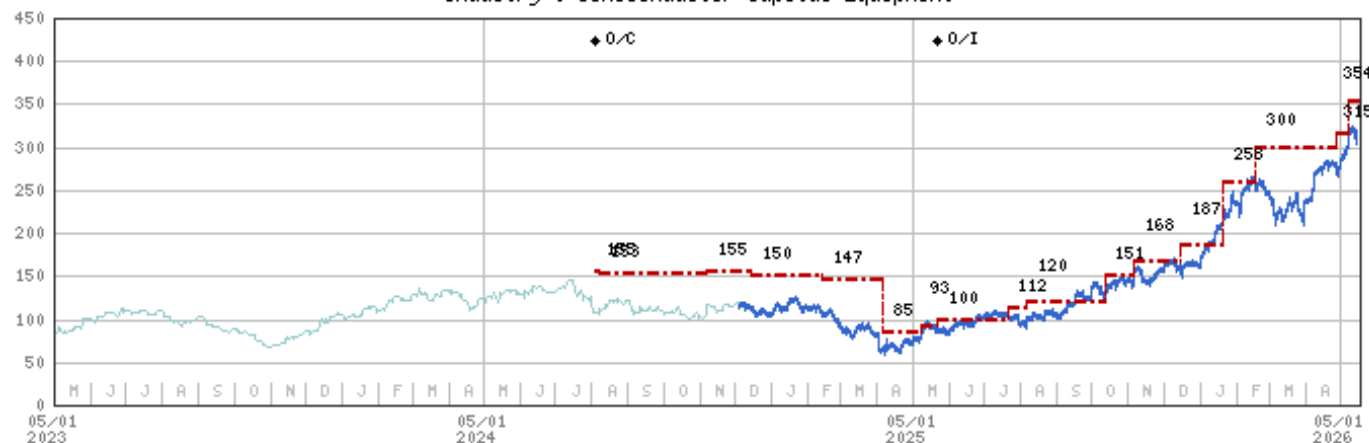
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

MKS Inc. (MKSI.O) - As of 05/17/26 GMT in USD
Industry : Semiconductor Capital Equipment



Stock Rating History: 8/4/24 : O/C; 5/22/25 : O/I

Price Target History: 8/4/24 : 155; 8/8/24 : 153; 11/7/24 : 155; 12/15/24 : 150; 2/13/25 : 147; 4/6/25 : 85; 5/8/25 : 93; 5/22/25 : 100; 7/21/25 : 112; 8/7/25 : 120; 10/12/25 : 151; 11/6/25 : 168; 12/15/25 : 187; 1/21/26 : 258; 2/18/26 : 300; 4/28/26 : 315; 5/7/26 : 354

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target — No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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INDUSTRY COVERAGE: Semiconductor Capital Equipment

COMPANY (TICKER)	RATING (AS OF)	PRICE* (05/15/2026)
Shane Brett		
Applied Materials Inc. (AMAT.O)	E (05/16/2026)	\$436.62
Camtek (CAMT.O)	E (12/01/2025)	\$168.35
KLA Corp (KLAC.O)	O (01/15/2026)	\$1,804.32

Lam Research Corp (LRCX.O)	O (05/16/2026)	\$284.72
MKS Inc. (MKS.O)	O (08/04/2024)	\$303.17
Nova Ltd (NVM.O)	E (12/01/2025)	\$509.15
Teradyne Inc (TER.O)	E (07/30/2025)	\$337.88

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.

INDUSTRY COVERAGE: Semiconductor Production Equipment

COMPANY (TICKER)	RATING (AS OF)	PRICE* (05/15/2026)
Suzune Tamura, CFA		
Advantest (6857.T)	O (01/17/2024)	¥26,360
DISCO (6146.T)	O (07/07/2023)	¥64,900
Ebara (6361.T)	O (12/04/2024)	¥5,533
JEOL (6951.T)	E (07/08/2025)	¥6,952
KOKUSAI ELECTRIC (6525.T)	O (12/19/2025)	¥6,590
Lasertec (6920.T)	U (07/28/2025)	¥38,410
Nikon (7731.T)	U (08/08/2023)	¥2,266
SCREEN Holdings (7735.T)	O (11/13/2025)	¥11,335
Tekscend Photomask (429A.T)	E (11/18/2025)	¥4,220
Tokyo Electron (8035.T)	O (12/19/2025)	¥50,290
Ulvac (6728.T)	E (01/17/2024)	¥9,486
Ushio (6925.T)	O (01/05/2026)	¥4,239

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* Historical prices are not split adjusted.